PHYSICS

1) If radius of a spherical bubble starts to increase with time t as r = 0.5t. What is the rate of change of volume of the bubble with time t = 4s ?

(1) 8π units/s

(2) 4π units/s

(3) 2π units/s

(4) π units/s

2) $\frac{d}{dx}\sqrt{\sin 2x}$ is equal to

(1) $(\sin 2x)^{-1/2}$

(2) $\cos 2x(\sin 2x)^{-1/2}$

(3) $2\cos 2x(\sin 2x)^{-1/2}$

(4) $\cos 2x(\sin 2x)^{1/2}$

3) If $x^3 + 3xy + y^3 = 1$ then correct option is/are

(A) $\left(\frac{dy}{dx}\right)_{(1,1)} = -1$

(B) $\left(\frac{dy}{dx}\right)_{(1,1)} = -2$

(C) $\left(\frac{dy}{dx}\right)_{(1,0)} = -1$

(D) $\left(\frac{dy}{dx}\right)_{(1,0)} = -3$

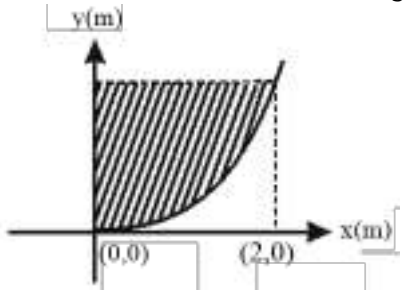
(1) A and B

(2) B and D

(3) A and C

(4) B and C

4) Find the area of shaded region between curved and y-axis is (Equation of curve is $y = x^2$) :



(1) $\frac{8}{3}$

(2) $\frac{16}{3}$

(3) $\frac{4}{3}$

(4) $\frac{2}{3}$

5) The velocity v of a particle is given by the equation $v = 6t^2 - 6t^3$ where v is in m/sec and t is time in seconds then :

(1) at t = 0, velocity is maximum

(2) at $t = \frac{2}{3}$, velocity is minimum

(3) minimum velocity is zero

(4) minimum velocity is – 2m/sec

6) Find component of $\vec{A} = \hat{i} + 2\hat{j} - 3\hat{k}$ in direction of $\vec{B} = 3\hat{i} + 2\hat{j}$:-

(1) $\frac{13}{7}\left(3\hat{i} + 2\hat{j}\right)$

(2) $\frac{7}{13}\left(3\hat{i} + 2\hat{j}\right)$

(3) $\frac{7}{\sqrt{14}}\left(\hat{i} + 2\hat{j} - 3\hat{k}\right)$

(4) $\frac{\sqrt{14}}{7}\left(\hat{i} + 2\hat{j} - 3\hat{k}\right)$

7) The value of the constant p such that the vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} - 3\hat{k}$ and $3\hat{i} + p\hat{j} + 5\hat{k}$ are coplanar is :

(1) zero

(2) 2

(3) –4

(4) –2

8) The maximum and minimum resultant of two forces acting at a point are 10N and 6N respectively. If each force is increased by 3N, find the resultant of new forces when acting at a point at an angle of 90° with each-other :-

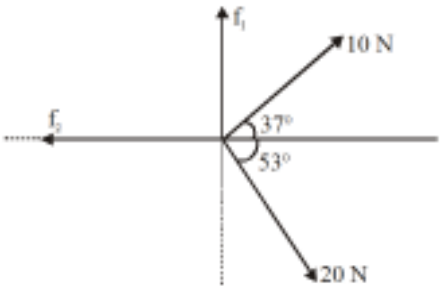
(1) $\sqrt{146}\,N$

(2) 11N

(3) $\sqrt{70}\,N$

(4) 8N

9) Four forces act on a particle as shown in the figure such that net force is zero. Then select the



correct statement :-

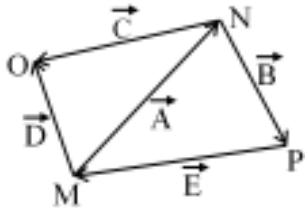
(1) Magnitude of $\vec{f}_1$ is 10N

(2) Magnitude of $\vec{f}_1$ is 20N

(3) Magnitude of $\vec{f}_2$ is 10N

(4) Magnitude of $\vec{f}_2$ is 30N

10) From figure, the correct relation is :-



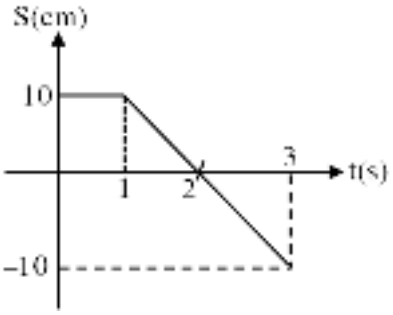
(1) $\vec{A}+\vec{B}+\vec{E}=\vec{0}$

(2) $\vec{C}-\vec{D}=-\vec{A}$

(3) $\vec{B}+\vec{E}-\vec{C}=-\vec{D}$

(4) All of the above

11) Find out impulse in interval 1s to 3s in the given displacement time graph. (m = 1kg)



(1) 10 kg-m/s

(2) 20 kg-m/s

(3) 30 kg-m/s

(4) Zero

12) A monkey is decending from the branch of a tree with a constant acceleration. If the breaking strength of the branch is 75% of weight of the monkey, the minimum acceleration with which the monkey can slide down without breaking the branch is -

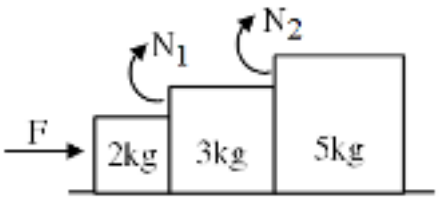
(1) g

(2) 3 g/4

(3) g/2

(4) g/4

13) Find out ratio of normal contact force N_1 and N_2 :



(1) $\frac{5}{3}$

(2) $\frac{8}{5}$

(3) $\frac{5}{9}$

(4) $\frac{3}{5}$

14) A ball is dropped on the floor from a height of 10 m. It rebound to a height of 2.5 m. If the ball is in contact with floor for 0.01 sec. What is the average acceleration during contact ?

(1) 700 m/sec^2

(2) 1400 m/sec^2

(3) 2100 m/sec^2

(4) 2800 m/sec^2

15) A Homogeneous Rod with length L is acted upon by two collinear forces F_1 and F_2 ($F_1 > F_2$) applied its ends and directed in opposite directions. What is tension force F at a cross-section P at a distance '□' from the end A.



(1) $\frac{(L-\ell)F_1 + \ell F_2}{L}$

(2) $\frac{L-\ell F_1}{L}$

(3) $\frac{\ell F_1}{L}$

(4) $\frac{\ell F_2}{L}$

16) A rope of length 8 m and linear density 0.5 kg/m is lying lengthwise on a horizontal smooth floor. It is pulled by a force of 12N. The tension at the midpoint would be :

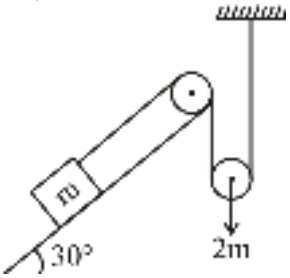
(1) 6 N

(2) 10 N

(3) 12 N

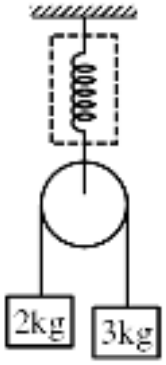
(4) 4 N

17) Consider the situation shown in figure. The acceleration of block of mass m is -



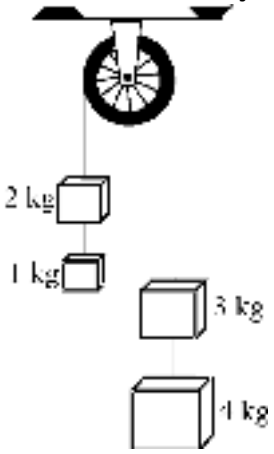
(1) g/3 up the plane
(2) g/3 down the plane
(3) g/2 up the plane
(4) g/2 down the plane

18) Find out reading of spring balance in given figure :



(1) 12 N
(2) 24 N
(3) 48 N
(4) 50 N

19) Find the acceleration of the system shown in figure.

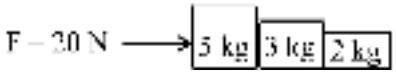


(g = 10 m/s^2) :-

(1) 2 m/s^2
(2) 4 m/s^2
(3) 8 m/s^2

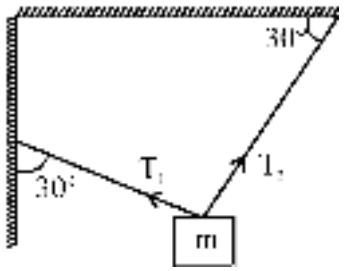
(4) 16 m/s^2

20) In the situation shown in fig. Force exerted by 3 kg block on 2 kg block is :-



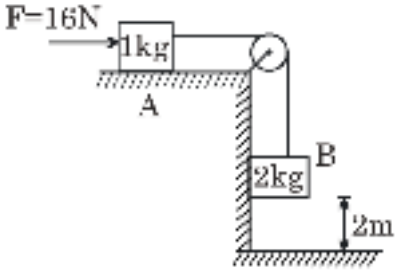
(1) 4 N
(2) 6 N
(3) 8 N
(4) 10 N

21) Calculate T_1 & T_2 :-



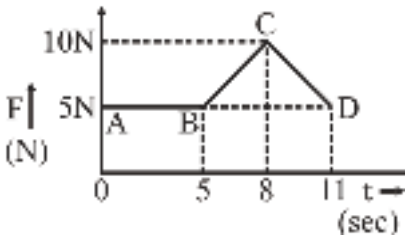
(1) $\frac{\sqrt{3}}{2}mg, \frac{mg}{2}$
(2) $mg, \frac{\sqrt{3}}{2}mg$
(3) $\frac{\sqrt{3}}{2}mg, \frac{\sqrt{3}}{2}mg$
(4) $\frac{mg}{2}, \frac{mg}{2}$

22) The system shown in the diagram is released from rest. Consider string segments as long. Neglect friction everywhere. Pulley and string are massless. If the time (in sec) after which block B reaches ground is t, find t ?



(1) $\sqrt{\frac{1}{5}}$ sec
(2) $\sqrt{\frac{2}{15}}$ sec
(3) $\sqrt{\frac{2}{5}}$ sec
(4) None

23) A force-time graph for a body is shown. Find the velocity of the body at the end of eleven second.



Assume the body to be initially at rest and mass of the body is 10 kg :-

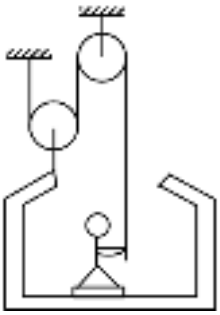
(1) 12 m/s

(2) 5 m/s

(3) 10 m/s

(4) 7 m/s

24) A man of 70 kg is standing on a weighing machine in a cage of 50 kg, as shown in the figure.



Find out tension in the string if system is in equilibrium :

(1) 300 N

(2) 400 N

(3) 600 N

(4) 900 N

25)

Five identical beads each of mass m connected to each other by means of five massless strings are suspended from a rigid support as shown in figure. The ratio of tensions in the strings 3 and 4 is:-



(1) 3 : 2

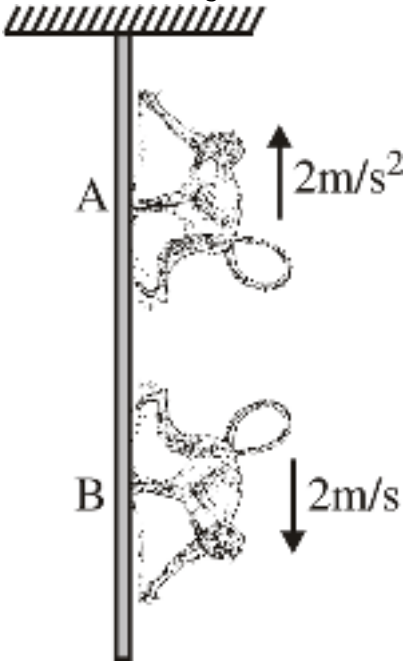
(2) 2 : 3

(3) 4 : 5

(4) 5 : 4

26) Two monkeys of masses 10 kg and 8 kg are moving along a vertical light rope, the former

climbing up with an acceleration of 2 m/s^2, while the latter coming down with a uniform velocity of 2 m/s. Find tension in the rope at the fixed support.



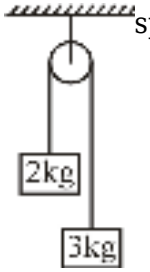
(1) 180 N

(2) 200 N

(3) 80 N

(4) 216 N

27) The adwood machine system shown in the figure is released from rest. At what time will the speed of the block of mass 3 kg be 6 m/s



(1) 3 sec

(2) 2 sec

(3) 1 sec

(4) 0.5 sec

28) A man is moving with velocity 5cm/sec towards a mirror. If the mirror is moving with velocity 5cm/sec towards the man then what will be the velocity of Image with respect to ground :-

(1) 0 cm/sec

(2) 5 cm/sec

(3) 10 cm/sec

(4) 15 cm/sec

29) A man 180 cm high stands infront of a plane mirror. His eyes are at a height of 170 cm from the floor. Then the minimum length of the plane mirror for him to see his full length image is :-

(1) 90 cm

(2) 180 cm

(3) 45 cm

(4) 360 cm

30) Rectilinear propagation of light means :

(1) Light is propagating very fast

(2) Light travels in a straight line

(3) Light is a wave

(4) None of the above

31)

Two plane mirror are inclined at an angle 80°. A ray of light is incident on one of mirror. The ray is undergo total deviation of :-

(1) 180°

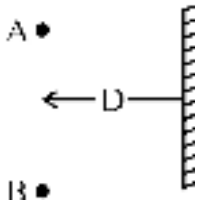
(2) 200°

(3) 90°

(4) 60°

32)

A beam of light starting from point A reflects from mirror M and reaches a point B as shown in adjoining figure in the same horizontal plane as A. For small angle of incidence i, AB is equal to :



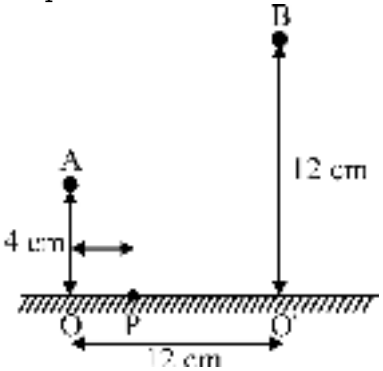
(1) iD

(2) 2iD

(3) 4iD

(4) iD/2

33) If a ray of light is travel from point A to point B after reflection from point P on plane mirror (given diagram) than find distance OP :-



(1) 6 cm

(2) 5 cm

(3) 3 cm

(4) 2 cm

34) An object is placed in front of a convex mirror of focal length 40 cm at a distance equal to focal length. Find position of image :-

(1) Infinity

(2) 20 cm

(3) 40 cm

(4) None of these

35) A point object is moving towards a concave mirror of focal length 25 cm. When it is at a distance of 20 cm from the mirror, its velocity is 5 cm/sec. Find velocity of image at that instant :-

(1) 10 cm/sec

(2) 125 cm/sec

(3) 25 cm/sec

(4) 5 cm/sec

36) Where should an object of height 5cm be placed In front of a concave mirror of focal length 10cm so that a real Image of height 1cm is obtained :-

(1) 48 cm

(2) 60 cm

(3) 15 cm

(4) 30 cm

37) Let the Transverse magnification produced by a spherical mirror be m. Then for the same position of object and mirror the longitudinal magnification will be :-

(1) m

(2) $\sqrt{m}$

(3) m^2

(4) 1/m

38)

A luminous point is moving at speed v_0 towards a spherical mirror, along its axis. Then the speed at which the image of this point object is moving is given by (with R = radius of curvature and u = object distance) :-

(1) $v_i = -v_0$

(2) $v_i = -v_0\left(\frac{R}{2u-R}\right)$

(3) $v_i = -v_0\left(\frac{2u-R}{R}\right)$

(4) $v_i = -v_0\left(\frac{R}{2u-R}\right)^2$

39) A square object of area 100 sq. cm is placed perpendicular to the principle axis of a concave mirror. If the lateral magnification of the mirror, for the above object position, is 0.4, then the area of the image will be :-

(1) 16 sq. cm

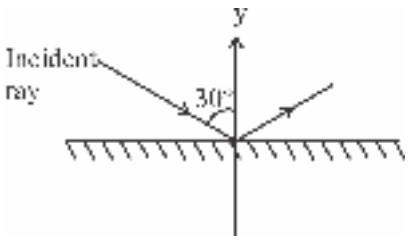
(2) 40 sq. cm

(3) 100 sq. cm

(4) 250 sq. cm

40)

By what angle mirror should be rotated to obtain reflected ray along + x-axis



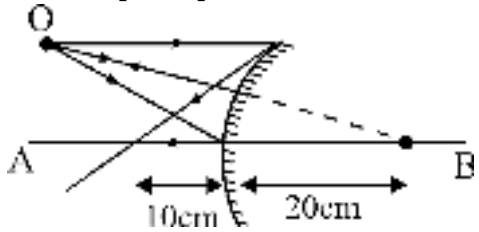
(1) 150°ACW

(2) 30°CW

(3) 60°CW

(4) 150°CW

41) In the figure shown a convex mirror of radius of curvature 20 cm is shown. An object O is placed in front of this mirror. Its ray diagram is shown. How many mistakes are there in the ray diagram (AB is its principal axis) :



(1) 3

(2) 2

(3) 1

(4) 0

42) An object of length 6 cm is placed perpendicular to the principal axis of a concave mirror of focal length f at a distance of 4f. The length of the image will be

(1) 2cm

(2) 12cm

(3) 4cm

(4) 1.2cm

43) When an object is placed at a distance of 25 cm from a mirror, the magnification is m_1. The object is moved 15 cm away with respect to the earlier position, magnification becomes m_2. If $m_1/m_2 = 4$, the focal length of the mirror is :

(1) 10 cm

(2) 30 cm

(3) 15 cm

(4) 20 cm

44) A point object is placed at a distance of 10 cm and its real image is formed at a distance of 20 cm from a concave mirror. If the object is moved by 0.1 cm towards the mirror, the image will shift by about :-

(1) 0.4 cm away from the mirror

(2) 0.4 cm towards the mirror

(3) 0.8 cm away from the mirror

(4) 0.8 cm towards the mirror

45) Sun subtends an angle of 0.5° at the pole of a concave mirror or radius of curvature 15 m. The diameter of the image of the sun formed by the mirror is

(1) 8.55 cm

(2) 7.55 cm

(3) 6.55 cm

(4) 5.55 cm

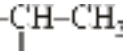
CHEMISTRY

1)

Which among the following is secondary radical?

(1) $CH_2=CH-$

(2) $(CH_3)_3C-$

(3) 

(4) $(CH_3)_2CHCH_2-$

2)

Which among the following homologous series has incorrect general formula ?

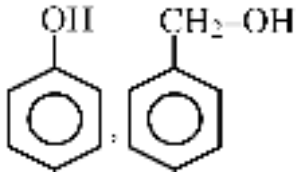
(1) Alkanol $C_nH_{2n+2}O$

(2) Alkanal $C_nH_{2n}O$

(3) Ester $C_nH_{2n}O_2$

(4) Alkenyne C_nH_{2n-2}

3) Which of following is pair of same homologous series ?

(1) 

(2) $CH_3-CH_2-CH_2-CH_2-NH_2$,
$CH_3-CH_2-NH-CH_3$

(3) CH_3-SH , CH_3-CH_2-OH

(4) CH_3-O-CH_3 , $CH_3-CH_2-O-CH_3$

4) How many olefinic bonds are present in Divinyl ketone -

(1) One

(2) Two

(3) Three

(4) Four

5)

Which of the following is not a primary alcohol:

(1) $CH_2 = CH - OH$

(2) $CH_2 = CH–CH_2–OH$

(3) $(CH_3)_2CH - OH$

(4) $(CH_3)_2CHCH_2OH$

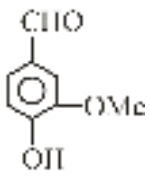
6) Find correct statement.

(1) the change in hybridisation, affects the electronegativity of carbon atom

(2) tetrahydrofuran is hetero cyclic compound

(3) the structural unit $(CH_3)_3CCH_2$ - is a neopentyl group.

(4) All are correct

7) Which functional group is not present in given compound? 

(1) Ether

(2) Aldehyde

(3) Alcohol

(4) Phenol

8) **Assertion :-** $CH_2 = CH - CH = CH_2$ is planar
Reason :- All carbon atom are sp^2 hybridised.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

9) Match the column

Column-I (Structure of compounds)		**Column-II (Hybridisation of C-atom)**	
(P)	CH_3–Cl	(A)	sp^3 and sp^2
(Q)	CH_3–CO–CH_3	(B)	sp^2
(R)	CH_3–C≡N	(C)	sp^3
(S)	$HCONH_2$	(D)	sp^3 and sp

Code :

	P	Q	R	S
(1)	C	A	D	B
(2)	D	C	B	A
(3)	C	D	A	B
(4)	A	B	C	D

(1) 1

(2) 2

(3) 3

(4) 4

10) Number of σ bond, π-bond and lone pair of electron respectively present in tropone are -

(1) 12, 4, 2

(2) 13, 4, 2

(3) 14, 4, 2

(4) 14, 3, 2

11) Neopentyl amine is:-

(1) Primary amine

(2) Secondary amine

(3) Tertiary amine

(4) None

12) The structure of isopropyl carbinol is :

(1) $(CH_3)_2CHOH$

(2) $CH_3 - CHOH - CH_2 - CH_3$

(3) $(CH_3)_2CHCH_2OH$

(4) $(CH_3)_3OH$

13) The IUPAC name of the compound
$CH_2 = CH - CH(CH_3)_2$ is :-

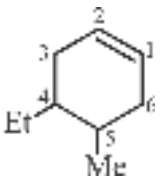
(1) 1,1-Dimethyl-2-propene

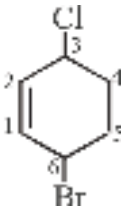
(2) 2-Vinylpropene

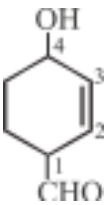
(3) 3-Methylbut-1-ene

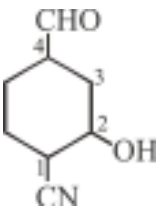
(4) 1-Isopropylethene

14) In which case IUPAC numbering is incorrect ?

(1) 

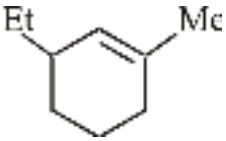
(2) 

(3) 

(4) 

15)

What is the IUPAC name of given compound ?



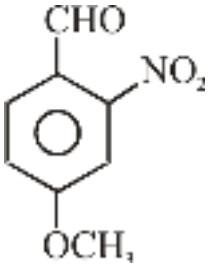
(1) 3-Ethyl-1-methylcyclohex-1-ene

(2) 1-Methyl-3-ethylcyclohex-1-ene

(3) 2-Methyl-5-ethylcyclohex-1-ene

(4) 2-Ethyl-6-methylcyclohex-1-ene

16) The correct IUPAC name of the compound is :-



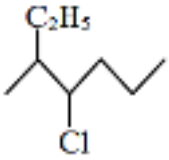
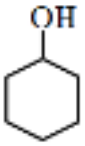
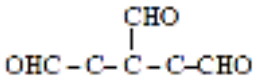
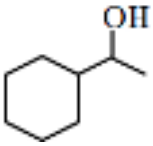
(1) 2-Formyl-5-methoxynitrobenzene

(2) 4-Formyl-3-nitroanisole

(3) 4-Methoxy-2-nitrobenzaldehyde

(4) 4-Methoxy-6-nitrobenzaldehyde

17)

	Column-I Compound		**Column-II** no. of C in parent carbon chain
A		P	6
B		Q	7
C		R	3
D		S	2

	A	B	C	D
(1)	P	Q	R	S
(2)	S	R	Q	P
(3)	Q	P	R	S
(4)	Q	R	S	P

(1) 1

(2) 2

(3) 3

(4) 4

18) The IUPAC name of the compound having the formula $CH{\equiv}C{-}CH{=}CH_2$ is :

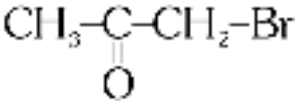
(1) But-1-yn-3-ene

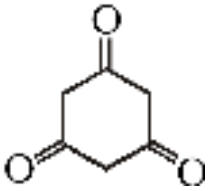
(2) Pent-1-yn-3-ene

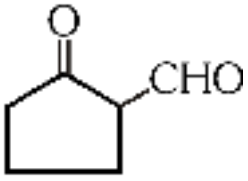
(3) But-1-en-3-yne

(4) But-3-en-1-yne

19) Which is correct IUPAC name :-

(1)  1 -Bromopropanone

(2)  1,3,5-Cyclohexanetrione

(3)  2-Keto cyclopentanecarbaldehyde

(4) All are correct

20) Which IUPAC name is incorrect ?

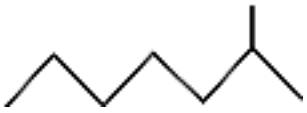
(1) 2-Methylcyclohexene

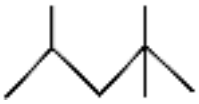
(2) Propan-3-ol

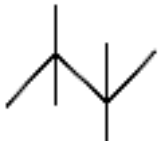
(3) 1-Amino-2-hydroxyethane

(4) All of these

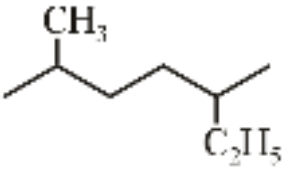
21) Structure of isooctane is-

(1) 

(2) 

(3) 

(4) 



22) IUPAC name of the Hydrocarbon is :-

(1) 2 - Ethyl - 5 - methylhexane

(2) 5 - Ethyl - 2 - methylhexane

(3) 2, 5 - Dimethylheptane

(4) 5- Ethyl - 2, 5 - dimethylpentane

23) **Assertion (A) :-** Rate of reaction increases with rise in temperature.
Reason (R) :- Activation energy of reaction decreases with rise in temperature.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is incorrect but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

24) **Assertion** : If in a zero order reaction, the concentration of the reactant is doubled, then half life period is also doubled.
Reason : For a zero order reaction, the rate of reaction is independent of initial concentration.

(1) Both assertion and reason are true and reason is a correct explanation of the assertion.

(2) Both assertion and reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

25) **Assertion :** The rate of hydrolysis reaction of ethyl acetate in acidic medium is affected by change in concentration of ethyl acetate, not by concentration of water.
Reason : For acidic hydrolysis of ethyl acetate, water is taken in large excess, therefore, concentration of water is not altered much during the reaction.

(1) Both **assertion** and **reason** are correct and **reason** is not the correct explanation of **assertion** .

(2) **Assertion** is correct but **reason** is incorrect.

(3) **Assertion** is incorrect but **reason** is correct.

(4) Both **assertion** and **reason** are correct and **reason** is the correct explanation of **assertion** .

26) Contact process is used in formation of sulphur trioxide $2SO_2 + O_2 \rightarrow 2SO_3$, the rate of reaction can be expressed as $-\frac{\Delta[O_2]}{\Delta t} = 8.0 \times 10^{-3} g L^{-1} s^{-1}$ then rate of disappearance of SO_2 will be

(1) 50.0×10^{-5} mol L^{-1} S^{-1}

(2) 3.75×10^{-4} mol L^{-1} S^{-1}

(3) 2.5×10^{-4} mol L^{-1} S^{-1}

(4) 4.12×10^{-4} mol L^{-1} S^{-1}

27) For the given reaction: $2A \rightarrow 3B + 4C$,if rate of disappearance of A is 4×10^{-2} Ms^{-1}. Find the rate of appearance of B at the same instant.

(1) 9×10^{-2}

(2) 10^{-2}

(3) 4.5×10^{-2}

(4) 6×10^{-2}

28) The following data is given for reaction between A and B :-

S.No.	**[A] mol. l^{-1}**	**[B] mol. l^{-1}**	**Rate mol. $l^{-1}sec^{-1}$**
I	1×10^{-2}	2×10^{-2}	2×10^{-4}
II	2×10^{-2}	2×10^{-2}	4×10^{-4}
III	2×10^{-2}	4×10^{-2}	8×10^{-4}

Which of the following are correct statements :-

(a) Rate constant of the reaction is 1 sec^{-1}

(b) Rate law of the reaction is k[A][B]

(c) Rate of reaction increases 4 times on doubling the concentration of both the reactant.

(1) a, b and c

(2) a and b

(3) b and c

(4) c alone

29) The reaction mechanism for the reaction
$2P \rightarrow R$ is as follows.

$$P \underset{K_2}{\overset{K_1}{\rightleftharpoons}} 2Q \text{ (Fast)}$$

$$2Q + P \xrightarrow{K_3} R \text{(Slow)}$$

The rate law for the main reaction ($P \rightarrow R$) is :

(1) $K_1[P][Q]$

(2) $K_1K_2[P]$

(3) $\frac{K_1K_3[P]^2}{K_2}$

(4) $K_1K_2[P]^2$

30) Incorrect statement is–

(1) Adsorption of NH_3 at Fe surface at high pressure is of zero order and at low pressure is of first order.

(2) Order determination may involve catalyst.

(3) Molecularity and order of reaction is not always same.

(4) Zero order reactions are completed in infinite time.

31) If half life of a zero order reaction having 0.02 M initial concentration is 200 sec then rate constant of reaction (in mol L/S) & time required for 100% reaction to be completed (in sec) is :

(1) 10^{-3}, 400

(2) 5×10^{-5}, 400

(3) 5×10^{-5}, 100

(4) 10^{-3}, 100

32) The rate constant of the reaction A → 2B is 1.0×10^{-3} mol L^{-1} min^{-1}, if the initial concentration of A is 1.0 mol L^{-1} what would be the concentration of B after 100 minutes.

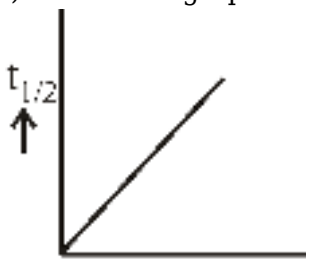
(1) 0.1 mol L^{-1}

(2) 0.2 mol L^{-1}

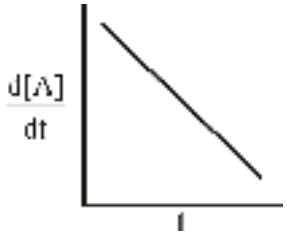
(3) 0.9 mol L^{-1}

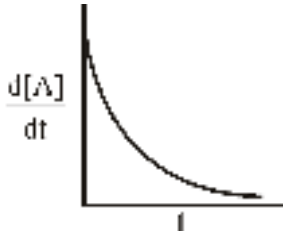
(4) 1.8 mol L^{-1}

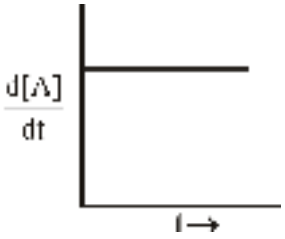
33) Consider a graph between half lite ($t_{1/2}$) and initial concentration of reactant (a) for reaction A →

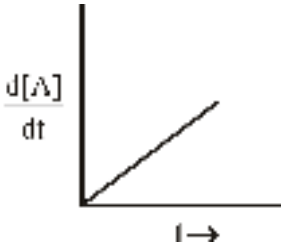


B So, graph between $-\frac{d[A]}{dt}$ and time will be :

(1) 

(2) 

(3) 

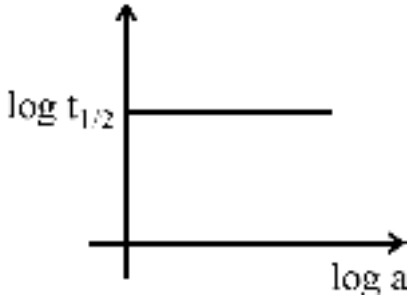
(4) 

34) If the rate constant of a reaction is 0.03 s^{-1}, how much time does it take for 7.2 mol L^{-1} concentration of the reactant to get reduced to 0.9 mol L^{-1} ? (Given : log 2 = 0.301)

(1) 69.3 s
(2) 23.1 s
(3) 210 s
(4) 21.0 s

35) If rate constant of a reaction is 2.31×10^{-3} s^{-1} and initial concentration of the reactant is 1.6 M then the rate of the reaction after second half life period will be

(1) 4.6×10^{-3} M s^{-1}
(2) 2.5×10^{-2} M s^{-1}
(3) 7.5×10^{-3} M s^{-1}
(4) 9.2×10^{-4} M s^{-1}

36) A graph between log $t_{1/2}$ and log a, 'a' being the initial concentration of reactant A is given, for reaction : A → product. 

If 75% of A is decomposed in 100 min. How long will it take for 90% of the compound to decompose.
[Given : log 2 = 0.3]

(1) 176.66 min.
(2) 156.66 min.
(3) 166.66 min.
(4) 150 min.

37) Which integrated equation is correct for the following Ist order reaction started with only A(g) in a closed rigid vessel.

$A(g) \rightarrow B(g) + C(g) + D(g)$
P_i = initial pressure;
P_t = total pressure at time t

(1) $K = \frac{2.303}{t} \log_{10} \left[\frac{P_i}{P_t}\right]$

(2) $K = \frac{2.303}{t} \log_{10} \left[\frac{P_t}{P_i}\right]$

(3) $K = \frac{2.303}{t} \log_{10} \left[\frac{2P_i}{3P_i - P_t}\right]$

(4) $K = \frac{2.303}{t} \log_{10} \left[\frac{3P_i}{2P_i - 3P_t}\right]$

38) If the half-life ($t_{1/2}$) for a first order reaction is 1 minutes, then the time required for 99.9% completion of the reaction is closest to:

(1) 2 minutes

(2) 4 minutes

(3) 5 minutes

(4) 10 minutes

39) An organic compound undergoes first order decompostion. The time taken for its decomposition to $\frac{1}{8}$ and $\frac{1}{10}$ of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of $\frac{t_{1/8}}{t_{1/10}}$?
[log 2 = 0.30]

(1) 0.09

(2) 0.9

(3) 9

(4) 90

40) For a certain reaction of order n the time for half change, t½ is given by $t½ = \frac{2-\sqrt{2}}{K}.(C_0)^{1/2}$ (Here K is rate constant, C_0 is initial concentration) then n is:-

(1) 1

(2) 2

(3) 0

(4) 0.5

41) Half - life of two samples of substance are 0.1 and 0.8 second. Their respective initial concentrations are 400 M and 50 M respectively initial. The order of reaction is :

(1) 0

(2) 2

(3) 1

(4) 4

42) Consider the following statements.
(A) The energy required to form activated complex is known as activation energy.
(B) Area under the curve for the plot of fraction of molecules versus kinetic energy of gas molecules, changes with change of temperature.
(C) In the Arrhenius equation the factor $e^{-Ea/RT}$ corresponds to fraction of molecules which collide with energy less than activation energy.
(D) Activation energy of reaction generally increases with increase in temperature.
The correct statement is :-

(1) A only

(2) B, C and D only

(3) D only

(4) A, B and C only

43) Match the Column I with Column II.

Column I		**Column II**	
A.	Catalyst alters the rate of reaction.	P	Cannot be fraction or zero
B.	Molecularity	Q	Proper orientation is not there.
C.	Second half-life of first order reaction.	R	By lowering the activation energy
D.	$e^{-E_a/RT}$	S	Is same as the first
E.	Energetically favourable reactions are sometimes slow.	T	Total probability is one.
F.	Area under the Maxwell - Boltzmann curve is constant.	U	Refers to the fraction of molecules with energy equal to or greater than threshold energy.

	A	B	C	D	E	F
(1)	R	P	S	U	Q	T
(2)	S	R	P	U	Q	T
(3)	P	R	S	Q	U	T
(4)	U	Q	T	S	R	P

(1) 1

(2) 2

(3) 3

(4) 4

44) The E_a of exothermic reaction X → Y is 40 KJ/mol. Heat of reaction is 30 KJ/mol. E_a for the reaction Y → X will be ?

(1) 10 KJ/mol

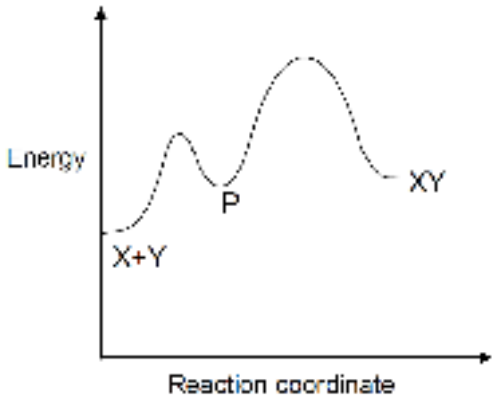
(2) 70 KJ/mol

(3) 40 KJ/mol

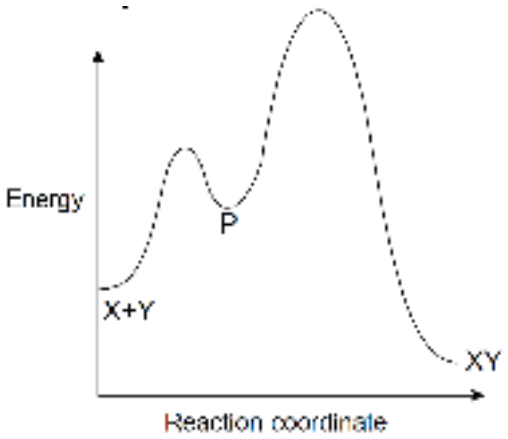
(4) 30 KJ/mol

45) For an exothermic chemical process occurring in two steps as follows:
(i) X + Y → P(slow)
(ii) P → XY(fast)
The energy diagram for the above process is best described by:

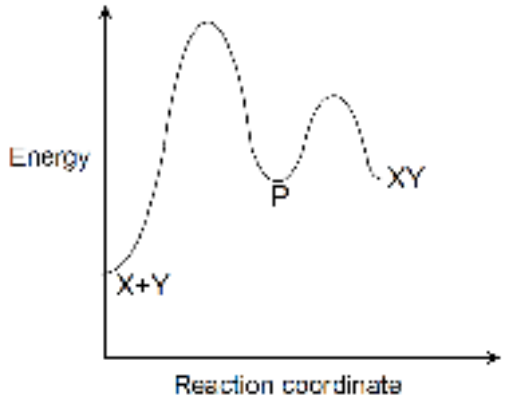
(1)



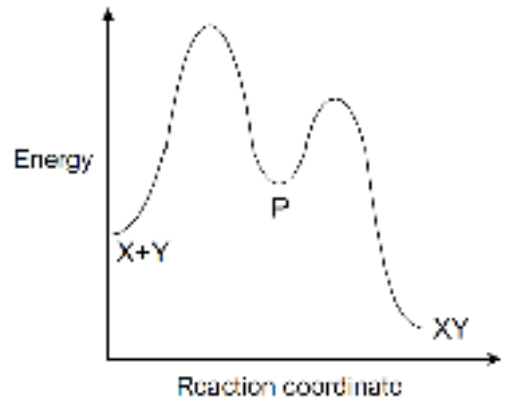
(2)



(3)



(4)



BIOLOGY

1) Which of the following method of birth control can cause excessive menstrual bleedings pain and inflammation of uterine wall ?

(1) Condom
(2) Cervical cap
(3) Oral contraceptive
(4) Intra uterine device (IUDs)

2) Which sequence is correct for male genital tract?

(1) Rete testis → Vas deferens → Vasa efferentia → Epididymis → Urethra
(2) Rete testis → Vas deferens→ Vasa efferentia → Seminal vesicle → Urethra
(3) Rete testis → Vasa efferentia → Vas deferens → Epididymis → Urethra
(4) Rete testis → Vasa efferentia → Epididymis → Vas deferens → Urethra

3) Seminal plasma in human is rich in :-

(1) Glucose, fructose, ribose
(2) Fructose, Calcium, certain enzymes
(3) Testosterone, AMH & Sperms
(4) Ribose, Testosterone, mucous

4) Which is not a natural method of contraception ?

(1) Coitus interruptus
(2) Periodic abstinence
(3) Vasectomy
(4) Lactational amenorrhoea

5) The first sign of growing foetus may be noticed by:-

(1) Movement of fetus
(2) Appearance of hair on head
(3) Listening to the heart sound carefully through the stethoscope
(4) Formation of limbs

6) The tertiary follicle is characterised by _____

(1) Presence of oocyte
(2) Presence of theca externa
(3) Presence of theca interna
(4) Presence of antrum

7) Which of the following is not a haploid cell?

(1) Secondary spermatocyte
(2) Spermatid
(3) Spermatogonium
(4) Sperm

8) The mammary glands of female undergo differentiation during pregnancy and starts producing milk towards the end of pregnancy by the process called :-

(1) lactation
(2) parturition
(3) ejaculation
(4) gestation

9) Which of the following is not included in the female external genitalia?

(1) Hymen
(2) Clitoris
(3) Infundibulum
(4) Labia majora

10) The funnel shaped part of fallopian tube is:

(1) isthmus
(2) ampulla
(3) infundibulum
(4) cervix

11) Which of the following male accessory glands helps in the lubrication of the penis?

(1) Seminal vesicles
(2) Prostate
(3) Bulbourethral glands
(4) Bartholin's glands

12) The first menstruation is called :

(1) Menopause
(2) Menarche
(3) Diapause
(4) Ovulation

13) The average duration of human pregnancy is:

(1) 28/29 days

(2) 270 – 290 days
(3) 24 weeks
(4) 3 months

14) Which of the following structures gets implanted in the uterus normally?

(1) 
(2) 
(3) 
(4) 

15) Which of the following hormones causes strong uterine contractions during parturition?

(1) Prolactin
(2) Oxytocin
(3) Relaxin
(4) Inhibin

16) Which of the following hormones is not secreted by placenta?

(1) Estrogen
(2) Progesterone
(3) hCG
(4) Oxytocin

17) Inner cell mass is part of:-

(1) morula
(2) Graafian follicle
(3) blastocyst
(4) zygote

18) Which hormone presence in urine of a pregnant female is helpful in detection of pregnancy?

(1) LH
(2) Progesterone

(3) FSH

(4) hCG

19) Birth canal is formed by :-

(1) Uterus and cervical canal

(2) Cervical canal and vagina

(3) Vagina and uterus

(4) Uterus, cervical canal and vagina

20) Read the following sentences and select correct option having all true statements:-
(a) During parturition, oxytocin induces intense contractions in the uterus
(b) After parturition oxytocin stimulate milk let-down.
(c) The labour pain during child birth is due to oxytocin
(d) Oxytocin is secreted from placenta and ovary

(1) b,c,d

(2) a,b,c

(3) c & d

(4) a & c

21) **Statement I :-** FSH acts on the sertoli cells and stimulates secretion of some factors which help in the process of spermiogenesis.
Statement II :- Due to high concentration of estrogen during folliculogenesis FSH increases more than the level of LH.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

22) Match List-I with List-II

List-I		**List-II**	
(A)	Embryo's heart formed	(I)	Fifth month of pregnancy
(B)	Foetus develops limbs & digits	(II)	End of II trimester
(C)	First movements of the foetus	(III)	End of II month of pregnancy
(D)	Eye lids separate	(IV)	After one month of pregnancy

(1) A→III ; B→I ; C→IV ; D→II

(2) A→IV ; B→III ; C→I ; D→II

(3) A→II ; B→IV ; C→I ; D→IV

(4) A→III ; B→IV ; C→II ; D→I

23) Find out the correct match from column I and column II

	Column-I		Column-II
A.	Morula	i.	Endometrium
B.	Blastocyst	ii.	Trophoblast
C.	Implantation	iii.	8-to 16 blastomeres
D.	Outer layer of blastocyst	iv.	Trophoblast and inner cell mass

(1) A – iv, B – iii, C – i, D – ii

(2) A – iv, B – ii, C – iii, D – i

(3) A – iii, B – i, C – iv, D – ii

(4) A – iii, B – iv, C – i, D – ii

24)

Arrange the following events of female reproductive cycle in (human) in sequence :-
(I) Secretion of FSH
(II) Formation of corpus luteum
(III) Growth of follicles
(IV) Ovulation
(V) Sudden increase in level of LH.

(VI) Formation of corpus albicans

(1) III → I → IV → II → VI → V

(2) I → III → V → IV → II → VI

(3) I → IV → III → V → II → VI

(4) II → I → III → IV → VI → V

25) Match the following and choose the correct answer :

(A)	Fertilization	(i)	Fundus of uterus
(B)	Implantation	(ii)	Ovary
(C)	Functional maturation of sperms	(iii)	Ampulla
(D)	Oogenesis	(iv)	Epididymis

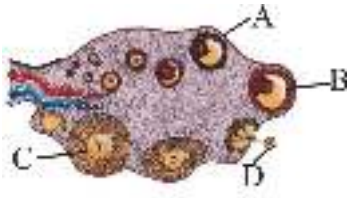
(1) A-(ii), B-(iii), C-(i), D-(iv)

(2) A-(ii), B-(i), C-(iv), D-(iii)

(3) A-(iii), B-(ii), C-(iv), D-(i)

(4) A-(iii), B-(i), C-(iv), D-(ii)

26) The figure below show section of ovary. Identification the structure which is essential for

maintenance of the endometrium :- 

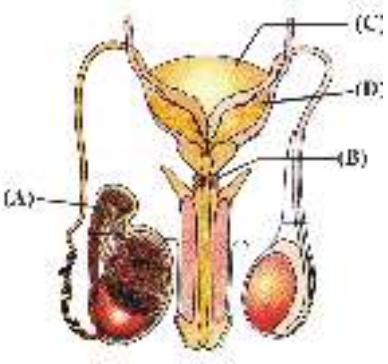
(1) A

(2) B

(3) C

(4) D

27) It is a diagrammatic sectional view of human male reproductive system. Which one of these is correctly identified in the options given, along with its correct function ?



	Structure	Function
(1)	D – Seminal vesicle	Store the sperms
(2)	B – Bulbourethral gland	Secrete fibrinogen
(3)	A – Epididymis	Functional maturation of Sperms
(4)	C – Urinary bladder	Storage of Urine and sperm

(1) 1

(2) 2

(3) 3

(4) 4

28) In the given four statements (A–D) select the options which includes all the **correct** ones only :-
(A) Tubectomy is the terminal method of conception in males
(B) Oral contraceptive pills prevents the ovulation from ovary.
(C) Copper releasing IUD's releases Cu ions, which suppress sperm motility and the fertilizing capacity of sperm.
(D) Condoms are the barrier method of contraception.

(1) Statements A, B, C correct

(2) Statements B, C, D correct

(3) Statements A, C, D correct

(4) Statements A, B, D correct

29) Which one of the following structure of sperm is mismatched ?

(1)	Acrosome	Contains sperm lysins that help in fertilisation
(2)	Nucleus	Contains haploid genome
(3)	Proximal centriole	Starts cleavage in zygote
(4)	Tail	Contains numerous mitochondria, which produce energy for movement of tail

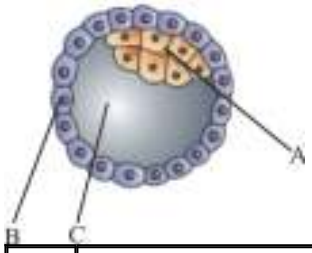
(1) 1
(2) 2
(3) 3
(4) 4

30) Identify the wrong one :

(1) 60% sperms must have normal shape and size for normal fertility
(2) 40% sperms must have normal shape and size for normal fertility
(3) Seminal plasma along with the sperm constitute the semen
(4) At puberty 60,000 – 80,000 primary follicles are left in each ovary

31)

In the given figure identify the parts labelled as A, B & C, and choose the correct option :-



	A	**B**	**C**
(1)	Inner cell mass	Trophoblast	Blastocoel
(2)	Trophoblast	Inner cell mass	Blastocoel
(3)	Blastomere	Trophoblast	Inner cell mass
(4)	Trophoblast	Blastocoel	Inner cell mass

(1) 1
(2) 2
(3) 3
(4) 4

32) The figure below show three stage (A, B, C) of human development. Select the option giving

correct identification : 

(1) A = Morula, B = Blastula, C = Zygote

(2) A = Blastocyst, B = Morula, C = Zygote

(3) A = Ovum, B = Blastocyst, C = Morula

(4) A = Blastocyst, B = Secondary oocyte, C = Morula

33) Match List-I with List-II :-

	List-I		**List-II**
(A)	Menstrual phase	(i)	Ovulation
(B)	Luteal phase	(ii)	Development of corpus luteum
(C)	Follicular phase	(iii)	Break down of endometrial living
(D)	Ovulatory phase	(iv)	LH surge

Choose the correct answer from options given below :-

(1) A-i, B-ii, C-iii, D-iv

(2) A-iii, B-ii, C-iv, D-i

(3) A-iii, B-i C-iv, D-ii

(4) A-ii, B-iii, C-i, D-iv

34) Which of the following statements are correct regarding female reproductive cycle :-
(A) In primate mammals cyclical changes during reproduction are called menstrual cycle.
(B) First menstrual cycle begins at puberty and is called menarche.
(C) During pregnancy all events of the menstrual cycle stop and there is no menstruation.
(D) Cyclic menstruation is an indicator of normal reproductive phase and extends between menarche to menopause.

(1) A and B only

(2) A, B and C only

(3) A, B, C and D

(4) A and D only

35) Which of the following statement is incorrect about the male reproductive system ?

(1) The testis are situated outside the abdominal cavity within a pouch called scrotum.

(2) The scrotum helps in maintaining the high temprature of the testes.

(3) The width of each testis is about 2 to 3 cm.

(4) Each testicular lobule contains one to three coiled seminiferous tubules.

36) Read the following statement.

(a) Each testis has about 350 testicular lobules.
(b) Sertoli cell provide nutrition to the leydig cell.
(c) Region outside seminiferous tubule called interstitial space.
(d) Government of India legalised MTP in 2017.
(e) "Family planning" programmes were initiated in 1951.
Choose the correct option :-

(1) a, b and d
(2) c and e
(3) b, c and e
(4) a, c and e

37) Which of the following group of hormones are produced in women only during pregnancy ?
Choose the most appropriate option.

(1) HCG, HPL, Inhibin
(2) HPL, Prolactin, Thyroxine
(3) HCG, HPL, Relaxin
(4) Relaxin, Cortisol, HCG

38) **Assertion (A) :-** All copulation does not leads to fertilisation and pregnancy.
Reason (R) :- Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region.

(1) Both A and R are true and R is the correct explanation of A.
(2) A is true but R is false.
(3) A is false but R is true.
(4) Both A and R are true and R is not the correct explanation of A.

39) How many of the followings are not secondary sex organs -
[Epididymis, Vagina, Bartholin gland, Testes, Penis, Uterus, Breasts, Oviduct, Ovary]

(1) 2
(2) 1
(3) 3
(4) 4

40) ___A___ start division and enter into ___B___ and get arrested into ___C___ .
Fill the blanks by choosing correct option -

(1) A – Primary spermatocyte, B – Prophase - I , C – Diplotene stage
(2) A – Primary oocyte, B – Meiosis- I , C – Diplotene stage
(3) A – Secondary oocyte, B – Meiosis - I , C – Metaphase
(4) A – Primary oocyte, B – Meiosis- II , C – Metaphase

41) Choose the correct option presenting correct match for the following columns.

	Column-I		Column-II
(a)	Endometrium	(i)	Cervical canal + vagina
(b)	Ovary	(ii)	IgA
(c)	Birth canal	(iii)	Primary reproductive organ
(d)	Colostrum	(iv)	Glandular layer
(e)	Oviducts	(v)	Fallopian tubes

(1) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii), (e)-(v)

(2) (a)-(v), (b)-(iii), (c)-(iv), (d)-(ii), (e)-(i)

(3) (a)-(iv), (b)-(iii), (c)-(i), (d)-(v), (e)-(ii)

(4) (a)-(iv), (b)-(v), (c)-(iii), (d)-(ii), (e)-(i)

42) At the puberty of a women they have 80 primary oocyte completing its meiosis-I, each of which converts into 2° oocyte and they have one child then how many I^{st} polar body formed during its life span?

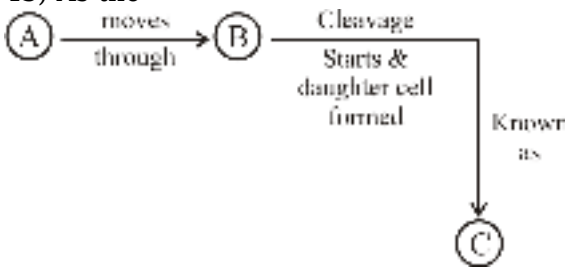
(1) One

(2) 40

(3) 80

(4) 79

43) As the



Find A, B, C :-

	A	B	C
(1)	Zygote	Ampulla	Morula
(2)	Zygote	Isthmus	Blastomeres
(3)	Egg	Isthmus	Blastula
(4)	Embryo	Isthmus	Blastomeres

(1) 1

(2) 2

(3) 3

(4) 4

44) (A) Surgical methods also called sterilisation.
(B) Sterilisation procedure in males - Tubectomy.

(C) Sterilisation procedure in feamles - Vasectomy.
(D) Surgical techniques are highly offective but their reversibility is very poor.
(E) Surgical methods also called as - periodic abstinence.
Choose incorrect points from above option :-

(1) Only A & D
(2) B, C & E only
(3) A, C, D & E
(4) B, C & E

45) Arrange the following events in the correct order as they take place during human reproduction :
(a) Insemination
(b) Gestation
(c) Fertilization
(d) Parturition
(e) Implantation

(1) (a), (b), (d), (c), (e)
(2) (a), (c), (e), (b), (d)
(3) (b), (d), (c), (a), (e)
(4) (a), (c), (e), (d), (b)

46) Citrus canker disease is caused by a species of :-

(1) Alternaria
(2) Erwinia
(3) Xanthomonas
(4) Pseudomonas

47) Heterocysts are related with :-

(1) Photosynthesis
(2) N_2 fixation
(3) Sexual reproduction
(4) All of these

48) Saprophytic protists having aggregation called plasmodium are :-

(1) Chrysophytes
(2) Dinoflagellates
(3) Slime moulds
(4) Protozoans

49) Red tide is caused by :-

(1) Rapid multiplication of red algae

(2) Rapid multiplization of euglenoids

(3) Rapid multiplication of slime molds

(4) Rapid multiplication of dinoflagellates

50) How many of the followings are photosynthetic.
Dinoflagellates, Diatoms, Euglenoids, Slime mould, Protozoa

(1) Four

(2) Three

(3) Five

(4) Two

51) Dinoflagellates are called as armoured algae because :-

(1) They have cell wall of cellulose

(2) Their cell wall is made up of plates

(3) They have transverse flagellum

(4) They are eukaryotic organisms

52) Chemoautotrophic bacteria that plays important role in nitrogen metabolism :-

(1) *Mycobacterium*

(2) *Nitrosomonas*

(3) *Rhodospirillum*

(4) *Clostridium*

53) Read the following statements carefully :-
(A) Some bacteria have resistence to certain antibiotics, this ability is provided by the extra genomic DNA of bacteria
(B) Mesosomes help in respiration and DNA replication
(C) Pili and fimbriae are the surface appendages of bacterial cell which play a role in motility.
(D) Glycocalyx, cell wall and cell membrane form the cell envelope of bacterial cell.
Which of the above statement(s) is/are correct and incorrect ?

	Correct statement	**Incorrect Statement**
(1)	A,B & D	C
(2)	A & B	C & D
(3)	B & D	A & C
(4)	B, C & D	A

(1) 1

(2) 2

(3) 3

(4) 4

54) Which one of the following is not the characteristic feature of cyanobacteria ?

(1) They form colonies

(2) They can fix atmospheric N_2

(3) They are eukaryotes

(4) They form blooms in polluted water bodies.

55) Which of the following are correct :
(A) Methanogens are an archaebacteria produce methane in marshy place
(B) Nostoc is blue green algae fix atmospheric N_2
(C) Chemosynthetic bacteria form cellulose from glucose
(D) Mycoplasma is a wall less organism that can survive without oxygen

(1) B and C

(2) A, B and C

(3) B, C and D

(4) A, B and D

56) Recognise the following figures and find the suitable options :-



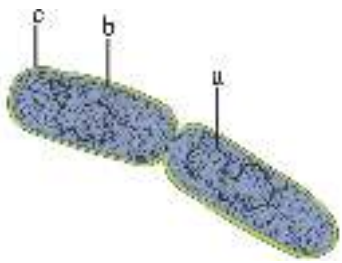
(i) A = *Paramoecium*, B = *Euglena*
(ii) Both 'A' and 'B' are Eukaryotes
(iii) A = *Euglena*, B = *Agaricus*
(iv) Both 'A' and 'B' are motile
(v) 'B' is Heterotrophic while 'A' is both autotrophic and some time heterotrophic.

(1) (i), (ii), (iv) & (v)

(2) (i), (iv) & (v)

(3) (ii), (iii) & (v)

(4) (ii), (iii), (iv) & (v)

57) 

Answer the following questions :-
(A) – Identify the figure
(B) – Identify the parts labelled a, b & c

	A	B

(1)	Dividing *Euglena*	a- Pellicle b- Cell membrane c- DNA
(2)	Dividing bacteria	a- DNA b- Cell membrane c- Cell wall
(3)	*Paramoecium*	a- DNA b- Cell membrane c-glycocalyx
(4)	Mycoplasma	a- Mesosome b- Cell membrane c- Cell wall

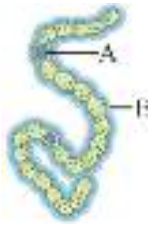
(1) 1

(2) 2

(3) 3

(4) 4

58) Given figure is of filamentous blue green alga *Nostoc*. Identify the parts marked as A and B and select the correct option.



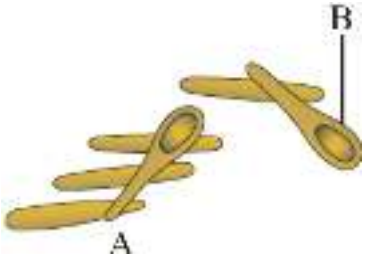
	A	**B**
(1)	Heterocyst	Mucilagenous sheath
(2)	Vegetative cell	Mucilagenous sheath
(3)	Trichomes	Cell wall
(4)	Mucilagenous sheath	Heterocyst

(1) 1

(2) 2

(3) 3

(4) 4

59) In the given diagram identify A and B and select correct answer. 

(1) A-Cocci, B-DNA

(2) A-Bacilli, B-Spore

(3) A-Spirilla, B-Spore

(4) A-Vibrio, B-Plasmid

60) 'Common mushroom we eat, moulds developed on our bread, ringworm, white spots on leaves of mustard etc. are due to actually :-

(1) Bacteria

(2) Algae

(3) Fungi

(4) All these characters are due to different types of organisms

61) Identify the correct match from the columns I, II and III :-

Column I		**Column II**		**Column III**	
1	*Albugo*	(a)	Ascomycetes	(i)	Obligate parasite
2	*Agaricus*	(b)	Deuteromycetes	(ii)	*Decomposer of litter,* parasites on potato
3	*Aspergillus*	(c)	Phycomycetes	(iii)	Laboratory Weeds
4	*Alternaria*	(d)	Basidiomycetes	(iv)	edible fungus

(1) 1-c-i, 2-a-iv, 3-d-iii, 4-b-ii

(2) 1-c-i, 2-d-iv, 3-b-iii, 4-a-ii

(3) 1-c-i, 2-a-iii, 3-d-iv, 4-b-ii

(4) 1-c-i, 2-d-iv, 3-a-iii, 4-b-ii

62) Choose the incorrect match from following :-

(1) *Mucor* - Phycomycetes

(2) *Puccinia* - Ascomycetes

(3) *Puff balls* - Basidiomycetes

(4) *Trichoderma* - Deuteromcycetes

63) Select the incorrect statement in the following:-

(1) Chrysophytes are found in fresh water as well as in marine environment.

(2) Dinoflagellates are mostly fresh water organism & photosynthetic

(3) Majority of Euglenoids are fresh water organism

(4) Slime moulds are saprophytic protist

64) What is true for archaebacteria ? Select correct answer according to codes.
1. Extreme halophiles
2. Methanogens
3. Presence of peptidoglycan in cell wall
4. Extreme thermophiles
Codes:

(1) 3 and 4 correct

(2) 2 and 3 correct

(3) 1, 3 and 4 correct

(4) 1, 2 and 4 correct

65) Choose the incorrect statement of following :-

(1) Ribosomes are the site of protein synthesis.

(2) Pili do not play a role in motility.

(3) Inclusion bodies are bounded by membrane.

(4) Cell wall prevent the bacterium from bursting and collapsing.

66) Bacterial structure and behaviour are respectively:

(1) Simple, Simple

(2) Complex, Simple

(3) Simple, Complex

(4) Complex, Complex

67) Heterocysts are found in :-

(1) Anabaena and *Nostoc*

(2) *Mucor* and *Chlorella*

(3) *Euglena* and *Paramoecium*

(4) *Lichens* and *Agaricus*

68) Which of the following are responsible for the production of methane in the gut of several ruminant animals :-

(1) Halophiles

(2) Acidophiles

(3) *E.coli*

(4) Methanogens

69) Both *Chlamydomonas* and *Chlorella* are autotrophic and both have cell wall whereas both *Paramoecium* and *Amoeba* are heterotrophic and both lack cell wall, then why R.H. Whittaker place all of them under the same kingdom ?

(1) Because all of them are single celled organisms.

(2) Because all of them have nuclear membrane and other membrane bound cell organelles

(3) Because all of them have cell membrane

(4) Both (1) & (2)

70) Match the column-I with column-II and select the correct option :-

Column-I		**Column-II**	
(A)	Monera	(I)	*Spirogyra, Solanum*
(B)	Protista	(II)	*Bacillus, Nostoc*
(C)	Plantae	(III)	*Euglena, Trypanosoma*
(D)	Fungi	(IV)	*Mucor, Rhizopus*

(1) A-III, B-II, C-I, D-IV

(2) A-II, B-III, C-I, D-IV

(3) A-II, B-IV, C-I, D-III

(4) A-II, B-I, C-IV, D-III

71)

Match Column-I with Column-II and select the correct option from the codes given below :-

Column-I		**Column-II**	
A	Chief producers in oceans	i	Euglenoids
B	Red tides	ii	Diatoms
C	Mixotrophic nutrition	iii	Slime moulds
D	Plasmodium	iv	Dinoflagellates

(1) A-(ii),B-(iv), C-(i), D-(iii)

(2) A-(ii),B-(iv), C-(iii), D-(i)

(3) A-(ii),B-(iii), C-(i), D-(iv)

(4) A-(i),B-(iv), C-(iii), D-(ii)

72) Protista is similar to plantae but differ from monera in having :-

(1) Mode of nutrition

(2) Cellular grade of organization

(3) Nuclear membrane

(4) Cell wall

73)

Given below are two statements :
One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : Deuteromycetes commonly known as Imperfect Fungi.
Reason (R) : In Deuteromycetes, sexual phase are not known.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

(3) Assertion is correct but Reason is not correct.

(4) Assertion is not correct but Reason is correct.

74) **Assertion :** In the five kingdom classification of Whittakar there is no mention of acellular organisms like virus and viroids.
Reason : Viruses and viroids did not find a place in classification since they are not truly living.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

75) Choose the wrong statement:

(1) Neurospora is used in the study of biochemical genetics

(2) Morels and truffles are poisonous mushrooms

(3) Yeast is unicellular and useful in fermentation

(4) Penicillium is multicellular and produces antibiotics

76) Carl Woese naturally clustered six kingdom into domain. Choose correct option for it

(1) Two domain for eukaryote and one domain for prokaryote

(2) One domain for eukaryote two domain for prokaryote

(3) Two domain for prokaryote and four domain for eukaryote

(4) Two domain for eukaryote and four domain for prokaryote

77) Which of the following statements are **true** about deuteromycetes ?
(a) Most of the members are decomposers
(b) Commonly known as imperfect fungi
(c) Most of the members are parasites
(d) Sexual phase of these fungi are known

(1) a & c

(2) a & d

(3) a & b

(4) c & d

78) Read the following statements carefully & find out correct statements for five kingdom system:
(A) Three kingdoms contain autotrophic organisms
(B) Only two kingdoms are of multicellular organisms
(C) Four kingdoms contain nuclear membrane
(D) All kingdoms contain chemosynthetic autotrophic organism

(1) C & D
(2) A & B
(3) A & C
(4) B & D

79) Select the correct combination of statements (A–D) regarding the characteristics of certain organisms.
(A) *Mycoplasma* lacks cell wall and can not survive without oxygen.
(B) Ascospores are 8 or 4 in number and endogenously formed.
(C) Fimbriae play important role in motility
(D) *Nostoc* and *Anabaena* are nitrogen fixing autotrophic bacteria found in coralloid root of cycas.
The correct statements are :-

(1) A & D
(2) C & D
(3) B & C
(4) B & D

80) Given below are two statements :-
Statement I :- Mycoplasma are the smallest living cells known and can survive without oxygen
Statement II :- Many Mycoplasma are pathogenic in animals and plants.
In the light of the above statements, Choose the most appropriate answer from the option given below :-

(1) Both statement I and statement II are incorrect
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statement I and statement II are correct.

81) **Statement-I :-** Instead of a cell wall, Euglenoides have a protein rich layer called pellicle which make their body flexible.
Statement-II :- Under favourable conditions, slime moulds form an aggregation called plasmodium.

(1) Both Statement I and Statement II are incorrect.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct.

82) **Statement-I :** Archaebacteria have complex cell wall and complex cell membrane. **Statement-II :** Archaebacteria can survive in extremely harsh habitats.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

83) Consider the following statements :-
I. In this group, the *Plasmodium* differentiates and forms fruiting bodies, bearing spores at their tips.
II. Spores possess true walls.
III. The spores are dispersed by air currents.
IV. The spores are extremely resistant and survive for many years even under adverse conditions.
The above statements are assigned to :-

(1) Euglenoids

(2) Slime moulds

(3) Dinoflagellates

(4) Chrysophytes

84) What is/are demerit(s) of two kingdom system of classification?

(1) It did not distinguish prokaryotes and eukaryotes

(2) It did not distinguish unicellular and multicellular organisms

(3) It did not separate photosynthetic and non-photosynthetic organisms

(4) All of the above

85) Find the incorrect statements from the following:- (A) Bacterial structure is very simple but very complex in behaviour (B) Bacteria show the most extensive metabolic diversity
(C) The vast majority of bacteria are autotrophs and a few are heterotrophs
(D) Bacterial cell wall is made up of lipoprotein

(1) Only A & B

(2) Only C & D

(3) Only A & D

(4) Only B & C

86) Read the following statements (A-D)
(A) In the five kingdom classification *Chlamydomonas* and *Chlorella* have been included in Protista
(B) *Entamoeba* commonly occurring bacterium in human intestine
(C) *Paramecium* and *Plasmodium* belong to the same kingdom as that of *Penicillium*
(D) Hydrogen donor in photosynthesis of sulphur bacteria is H_2S.
How many of the above statements are incorrect?

(1) Four

(2) One

(3) Two

(4) Three

87)

How many organism in the list given below are photosynthetic :-

Chlorobium, Clostridium, Mycobacterium, Nitrobacter, Nitrococus, Euglena, Albugo, E.coli, Nostoc, Anabaena.

(1) Six

(2) Eight

(3) Four

(4) Five

88) Consider the following organisms
Nitrobacter, Nostoc, Azotobacter, Nitrosomonas, Lactobacillus, Rhodospirillum, Beijernekia, Anabaena, Frankia.
How many of the above organisms can fix atmospheric nitrogen?

(1) Six

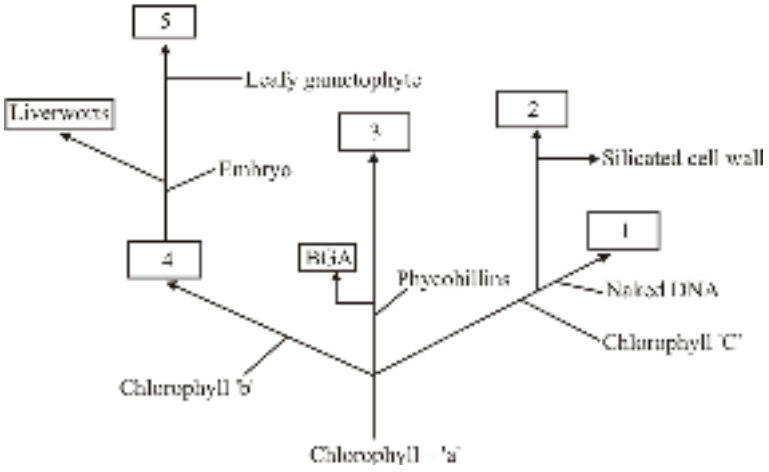
(2) Five

(3) Eight

(4) Seven

89)

A cladogram showing evolution of plants given below. Study it carefully and give the answer of following question.



Number '2' representing the which group of plants:-

(1) Dinoflagellates

(2) Moss

(3) Diatom

(4) Red algae

90) How many of the following statements are true for fungi ?
(A) Members of kingdom fungi show great diversity in morphology and habitat
(B) Most of fungi are saprophytic in their mode of nutrition
(C) Asexual cycle in fungi involve three steps
(D) Morels and truffles are edible mushroom and are considered delicacies
(E) *Albugo* is the parasitic fungi on mustard and wheat

(F) Members of ascomycetes can be multicellular or unicellular

(1) Two
(2) Four
(3) Three
(4) Five

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	2	3	2	3	2	3	1	1	4	4	4	2	3	1	1	1	3	2	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	3	4	2	1	2	1	4	1	2	2	2	3	2	2	2	3	4	1	2
Q.	41	42	43	44	45															
A.	2	1	4	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	4	4	2	3	4	3	1	1	3	1	3	3	2	1	3	3	3	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	3	2	2	4	1	4	3	3	4	2	2	3	1	4	3	3	4	2	4
Q.	86	87	88	89	90															
A.	2	1	1	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	4	2	3	3	4	3	1	3	3	3	2	2	3	2	4	3	4	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	4	2	4	3	3	2	4	2	1	4	2	3	2	2	3	1	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	2	2	2	3	2	3	4	2	2	2	1	3	4	3	2	1	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	2	2	4	3	3	1	4	4	2	1	3	1	1	2	2	3	3	4	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	2	4	2	3	3	1	3	3										

SOLUTIONS

PHYSICS

1) $V = \frac{4}{3}\pi r^3$

$\frac{dV}{dt} = \frac{4}{3}\pi \cdot 3r^2 \frac{dr}{dt}$

$= 4\pi\left(\frac{t}{2}\right)^2 \cdot \frac{1}{2} = \frac{t^2}{2}\pi = 8\pi$

2) $\frac{d}{dx}\sqrt{\sin 2x}$ from chain rule

$\Rightarrow \frac{1}{2}(\sin 2x)^{-1/2}.(\cos 2x)(2)$

$\Rightarrow (\cos 2x)(\sin 2x)^{-1/2}$

3) $\frac{d}{dx}\left(x^3 + 3xy + y^3\right) = \frac{d}{dx}(1)$

$3x^2 + 3x.\frac{dy}{dx} + 3y + 3y^2.\frac{dy}{dx} = 0$

$\frac{-3x^2 - 3y}{3y^2 + 3x} = \frac{dy}{dx}$

at (1, 1)

$\frac{dy}{dx} = \frac{-6}{6} = -1$

at (1, 0)

$\frac{dy}{dx} = \frac{-3}{3} = -1$

4) **Question Explanation:** Area bounded by y-axis.

Concept: Integration along y-axis.

Solution: The area is bounded by the y-axis from y = 0 to y = 4 (since x = 2, $y = 2^2 = 4$).

Express x in terms of y : $x = \sqrt{y}$

Integrate: Area $= \int_0^4 \sqrt{y}dy = \left[\frac{2}{3}y^{3/2}\right]_0^4$.

Calculation: $\frac{2}{3}(4)^{3/2} = \frac{2}{3}(8) = \frac{16}{3}$

Final Answer: Option (2)

5) $v = 6t^2 - 6t^3 \Rightarrow \frac{dv}{dt} = 12t - 18t^2$

For max. and min., $\frac{dv}{dt} = 0$

$\Rightarrow 12t - 18t^2 = 0$

$6t(2 - 3t) = 0$

t = 0, t = 2/3

$\frac{d^2v}{dt^2} = 12 - 3t$

at t = 0, $\frac{d^2v}{dt^2} = 12$

and $t = \frac{2}{3}, \frac{d^2v}{dt^2} = -12$

thus v is max at t = $\frac{2}{3}$ and min. at t = 0

6) **Explain Question:** Find the vector component (projection) of $\vec{A}$ in the direction of $\vec{B}$.

Concept: Projection of a vector on another vector (Application of dot product)

Solution: Vector projection formula:

Component of $\vec{A}$ along $\vec{B}$ $= \frac{\vec{A} \cdot \vec{B}}{\left|\vec{B}\right|^2}\vec{B}$

Given

$\vec{A} = \hat{i} + 2\hat{j} - 3\hat{k}, \vec{B} = 3\hat{i} + 2\hat{j}$

Dot product : $\vec{A} \cdot \vec{B} = (1)(3) + (2)(2) + (-3)(0) = 3 + 4 = 7$

Magnitude squared of $\vec{B}$

$\left|\vec{B}\right|^2 = 3^2 + 2^2 = 9 + 4 = 13$

Projection vector : Component of $\vec{A}$ along $\vec{B} = \frac{7}{13}\left(3\hat{i} + 2\hat{j}\right)$

Final answer: Option (2)

7) $(\vec{P} \times \vec{Q}).\vec{R} = 0$ (coplanar).

8) $F_1 + F_2 = 10$

$F_1 - F_2 = 6 \Rightarrow F_1 = 8, F_2 = 2$

New $F_1^1 = 8 + 3 = 11$

$F_2^1 = 2 + 3 = 5$

$R = \sqrt{11^2 + 5^2 + 2 \times 11 \times 5\cos 90^\circ}$

$= \sqrt{121 + 25} = \sqrt{146}$

9) **Explain question:**

Select correct statement for magnitude of vector.

Concept:

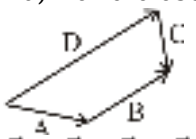
Resolution of vectors.

Solution:

$f_1 = -10 \sin 37° + 20 \sin 53°$
$= -6 + 16 = 10$ N
$f_2 = 10 \cos 37° + 20 \cos 53°$
$= 8 + 12 = 20$ N

Final answer: option (1)

10) For a closed polygon



$\vec{D} = \vec{A} + \vec{B} + \vec{C}$

11) $\vec{I} = \Delta\vec{p} = m\Delta\vec{v}$
$\Delta\vec{v} = 0$ (v = constant)
$\vec{I} = 0$

12)



$$a = \frac{mg - T}{m}$$

$mg - T = ma$

$$mg - \frac{3}{4}mg = ma$$

$$a = \frac{g}{4}\ m/sec^2$$

13) $N_1 = (3 + 5)\ a$
$N_2 = (5)\ a$

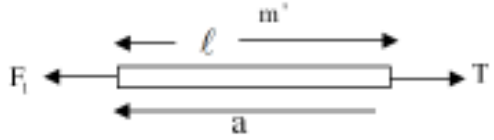
$$\frac{N_1}{N_2} = \frac{8}{5}$$

14) $$\left|\overrightarrow{a_{avg}}\right| = \frac{\overrightarrow{\Delta u}}{\Delta t} = \frac{v_f + v_i}{\Delta t} = \frac{\sqrt{2gh_f} + \sqrt{2gh_i}}{\Delta t}$$

$$= \frac{\sqrt{2(9.8) \times 2.5} + \sqrt{2(9.8) \times 10}}{0.01} = 2100 m/sec^2$$

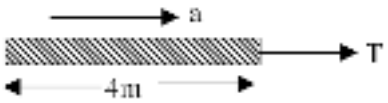
15) $$a = \frac{F_1 - F_2}{M}$$

$F_1 - T = m'a$



$$F_1 - T = \left(\frac{M}{L}\ell\right)\left(\frac{F_1 - F_2}{M}\right)$$
$$F_1 - T = (F_1 - F_2)\frac{\ell}{L}$$
$$T = F_1 - (F_1 - F_2)\frac{\ell}{L}$$
$$= \frac{F_1(L-\ell) + \ell F_2}{L}$$

16) mass of rope = 8 × 0.5 = 4 kg



$$a = \frac{12}{4} = 3 \text{ m/sec}^2$$
$$T = ma$$
$$= \frac{4}{2} \times 3 = 6N$$

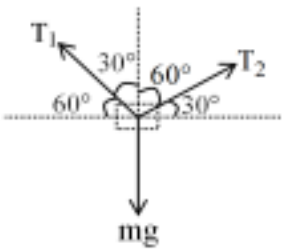
17) Along the plane
$T - mg \sin 30° = ma$
$$T - \frac{mg}{2} = ma \quad \text{....(1)}$$
For block of mass 2m
$2mg - 2T = 2m(a/2)$ ....(2)
By solving both eq. $a = g/3$
 acceleration of mass m = a
 acceleration of mass 2m = a/2

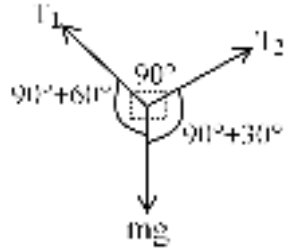
18) $$T = \frac{2(30) + 3(20)}{5} = \frac{120}{5} = 24 \text{ N}$$
Reading (2T) = 48 N

19) $$a = \frac{[(4+3) - (2+1)](g)}{(4+3) + (2+1)}$$
$$a = \frac{4(g)}{10} = 4 \text{ m/s}^2$$

20) $$a = \frac{20}{5+3+2} = 2\text{m/s}^2$$
Force exerted by 3 kg block on 2 kg block = 2(2) = 4N

21)





Apply LAMI's Theorem

$$\frac{mg}{\sin 90^o} = \frac{T_1}{\sin(90+30)^o} = \frac{T_2}{\sin(90+60)^o}$$

$$\frac{mg}{1} = \frac{T_1}{\sin 60^o} = \frac{T_2}{\sin 30^o}$$

$T_1 = mg \sin 60°$

$\Rightarrow T_1 = mg\frac{\sqrt{3}}{2}$

$T_2 = mg \sin 30° = mg/2.$

22) String will slack (T = 0), hence acceleration of B is g (10 m/s^2)

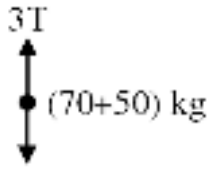
u = 0, s = 2 m, a = 10 m/s^2

$2 = \frac{1}{2} \times 10 \times t^2 \Rightarrow t = \sqrt{\frac{2}{5}}$

23) Δp = area of f–t graph

10(v) = 70

v = 7 m/s



24) 1200

3T = 1200

$\boxed{T = 400}$

25) $\frac{T_3}{T_4} = \frac{3mg}{2mg} = \frac{3}{2}$

26)

Tension = T – mg = ma

T = m(g + a)

T = 10(10 + 2)

T = 10 × 12 = 120

Upper support

120 + 80 = 200

(80) weight of latter monkey

27) **Question Explanation :** An Atwood machine has two masses 2 kg and 3 kg connected by a light string over a frictionless pulley. The system is released from rest. We have to find the time when the 3 kg block reaches speed 6 m/s.

Concept: Atwood machine-Newton second law and equations of motion.

Solution:
Acceleration of system

$$a = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g = \left(\frac{3-2}{5}\right) \times 10 = 2 m/s^2$$

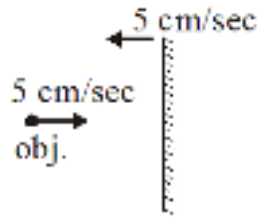
Velocity at time t

$v = u + at$

$$6 = 0 + 2 \times t \Rightarrow t = \frac{6}{2} = 3 \text{ sec.}$$

Final answer: Option (1)

28) $v_I = 2v_m - v_0$



$= 2(-5) - 5$
$= -15$

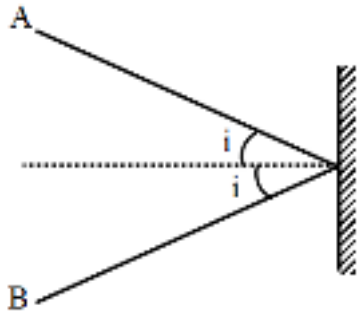
29) Minimum length of required mirror should be half of height of man.

30) **Explaining :** To find the means of given word
Concept : This question is based on
Light always propagate in straight line
Solution : Light always propagate in straight line
Light has mature : particle & wave
Final Answer : (2)

31)

$\delta = 360 - 2\theta$
$= 360 - 2 \times 80$

= 200°

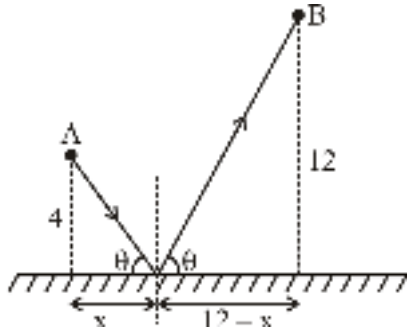


32)

$$\text{Angle} = \frac{\text{Arc}}{\text{Radius}}$$

$$2i = \frac{\text{Arc}}{R}$$

Arc = 2iD



33)

$$\tan\theta = \frac{4}{x} = \frac{12}{12-x}$$

48 – 4x = 12x

16x = 48

x = 3 cm

34)

Explanation : Object is placed in front of a convex mirror of focal length 40 cm; object distance equals focal length, so u = –40 cm and f = +40 cm. Required: position of image.

Concept : This question is based on mirror formula (Convex mirror).

Solution :

$$\frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{u}$$

Substitute f = +40 and u = –40.

$$\frac{1}{40} = \frac{1}{v} - \frac{1}{40}$$

$$\frac{1}{v} = \frac{1}{40} + \frac{1}{40} = \frac{2}{40} = \frac{1}{20}$$

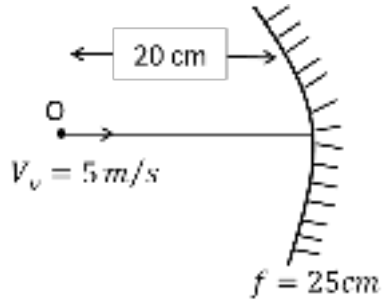
v = +20 cm (image behind mirror).

Final Answer : Option (2)

35) **Question Explanation:**
to calculate the velocity of image if object is moving with 5m/s towards the mirror (concave)

Concept:
Velocity of image in spherical mirrors.

Solution:



$u = -20cm$

$f = -25cm$ $\quad M_T = \frac{f}{f-u} = \frac{-25}{-25-(-20)} = \frac{-25}{-5} \quad M_T = 5$

$\vec{V_I} = -M_T^2 \ \vec{V_O}$

$\vec{V_I} = -(5)^2(5) = -125cm/s$

Final Answer: (2).

36)

Explain the Question : Object height = 5 cm, image height = 1 cm (real image), concave mirror focal length = 10 cm. Find object distance.

Concept : This question is based on image formation by a concave mirror.

Formula : $m = \frac{h_i}{h_o} = \frac{f}{f-u}$

Solution : Magnification $m = \frac{h_i}{h_o} = -\frac{1}{5}$

$m = \frac{h_i}{h_o} = \frac{f}{f-u}$

$\frac{-1}{5} = \frac{-10}{-10-u}$

$10 + u = -50$

$u = -60$ cm

Final answer : Option (2)

37)

Lateral magnification m = $-\frac{v}{u}$

Longitudinal magnification = $\left(\frac{v}{u}\right)^2$ = m^2

38)

Question Explanation: Speed of image in mirrors.
Concept: Longitudinal magnification.

Solution: The relationship between object velocity ($v_o = \frac{du}{dt}$) and image velocity ($v_i = \frac{dv}{dt}$) is derived from the mirror formula by differentiation:

$$v_i = -\left(\frac{v}{u}\right)^2 v_o$$

Using the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{2}{R}$, we find $\frac{v}{u} = \frac{R}{2u-R}$. Substituting this gives:

$$v_i = -v_o \left(\frac{R}{2u - R}\right)^2$$

Final Answer: Option (4)

39) **Question Explanation :**
To find the area of image of a square object in concave mirror.

Concept :
Areal magnefication in concave mirror.

Formula :
Area of image
$AI = (m)^2 . A_0$

Solution/Calculation :
Given m = 0.4
A_0 = 100.39 cm
I_I = 100.39 cm
$A_I = (0.4)^2 \times 100 = 0.16 \times 100 = 1639$ cm

Answer : 1
Level : Easy

40)

$m = \frac{h_i}{h_0} = \frac{f}{f-u}$

$\Rightarrow \frac{-3}{2} =$ ×

⇒ f = – 12 cm

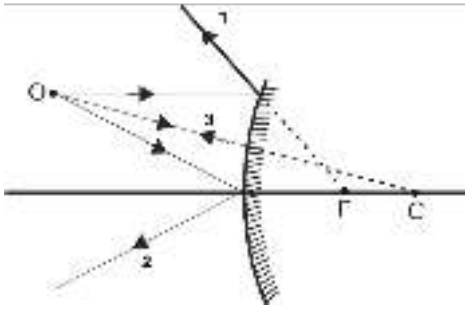
41) **Explain question:**

A convex mirror with radius of curvature 20 cm is shown with a ray diagram. We are to find how many mistakes are present in the diagram.

Concept:
Ray rules for convex mirror:

A. Incident ray parallel to principal axis reflects as if from focus.
B. Incident ray directed toward centre reflects back on itself.
C. The ray incident at the pole will be reflected at the same angle from principle axis.

The correct Diagram will be as follows



Solution:
Upon inspecting the given diagram:
1. Reflected ray not diverging correctly from the virtual focus.
2. Ray aimed at centre not reflecting back properly.
3. The ray aimed at the pole is not reflecting according to the rule.
Total mistakes = 2

Final Answer: Option (2)

42) $\frac{I}{O} = \frac{f}{f-u} \Rightarrow \frac{I}{+6} = \frac{-f}{-f-(-4f)}$
$\Rightarrow I = -2cm$

43)

$m_1 = \frac{f}{f+25}$
$m_2 = \frac{f}{f+40}$
$4 = \frac{f+40}{f+25}$
4f + 100 = f + 40
3f = – 60
f = –20 cm

44) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ here u = –10 cm, v = – 20 cm
differentiate it $-\frac{1}{v^2}\frac{dv}{dt} - \frac{1}{u^2}\frac{du}{dt} = 0$

$dv = -\frac{v^2}{u^2}du$, du = 0.1 cm towards mirror

$dv = -\left(\frac{-20}{-10}\right)^2 (0.1) = -0.4cm$

So dv = 0.4 cm away from mirror

45)

Image of sun at focus

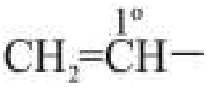
Diameter = $f\theta = \frac{R}{2} \times \left(0.5^\circ \times \frac{\pi}{180}\right)$

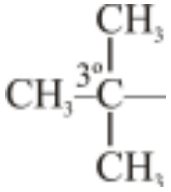
$= \frac{15}{2} \times \frac{0.5 \times 3.14}{180} = 6.55 \times 10^{-2}$ m

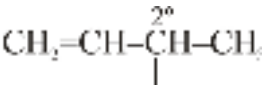
= 6.55 cm

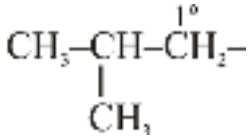
CHEMISTRY

46)

(1) 

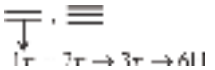
(2) 

(3) 

(4) 

* Secondary radical means free valency on 2°C

47)

Alkenyne → 

General formula of Alkenyne → $\boxed{C_nH_{2n-4}}$

48) Members of Homologous series have same functional group and difference of $-CH_2$ or $2(CH_2)$ or $3(CH_2)$ or so on.

49) 

C = C double bond are two.

50) 1. $\overset{(1^\circ)}{CH_2=CH-OH}$

$CH_2 = CH - CH_2 - OH$

(1°)

2. $(CH_3)_2CH-OH(2^\circ)$

$(CH_3)_2CH-CH_2-OH$

(1°)

51)  → Heterocyclic compound

THF

52)

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53)

$CH_2 = CH - CH = CH_2$ is planar due to all carbon are sp^2 hybridised.

54)

(P) $\overset{sp^3}{CH_3}-Cl$

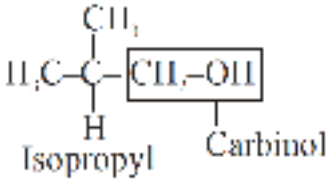
(Q) $\overset{sp^3}{CH_3}-\overset{sp^2}{CO}-\overset{sp^3}{CH_3}$

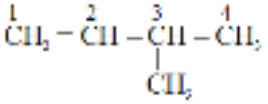
(R) $\overset{sp^3}{CH_3}-\overset{sp}{C}\equiv N$

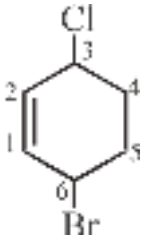
(S) $H\overset{sp^2}{C}ONH_2$

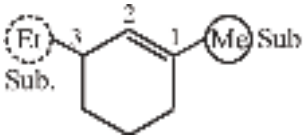
55) NCERT 11th Pg. # 331

56) $CH_3-\overset{CH_3}{\underset{CH_3}{C}}-CH_2-NH_2$ (1° Amine)

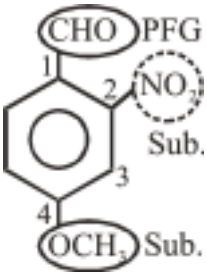
57) 

58) 

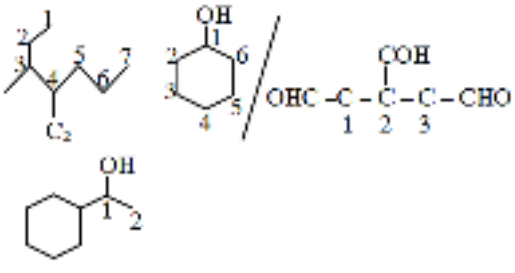
59) 
is correct IUPAC numbering.

60) 

3-Ethyl-1-methylcyclohex-1-ene

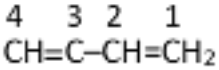
61) 

4-methoxy-2-nitrobenzaldehyde

62) 

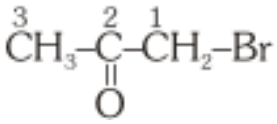
63) **Question Explanaion :**
The question asks that the UPAC name of compound having formula CH ≡ C – CH = CH_2
Concept:
This question is based on IUPAC nomenclature of hydrocarbon.

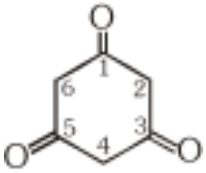
Solution :

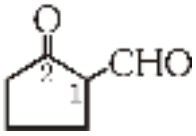


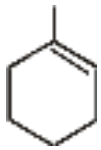
Selection of PCC : Including multiple bond the longest carbon cahain have 4 carbon.
By the numbering from both side multiple bond gets locant 1, 3 and 1, 3. So, we use alphabetically order and give priority do ene over yne.
Therefore numbering is done from right side.
IUPAC Name is :
1-Butene-3-yne
or
But-1-en-3-yne Here multiple bonds are present at symmetrical positions, so alkene (C=C) is given priority over alkyne (C≡C) while numbering. General name: Alkenyne

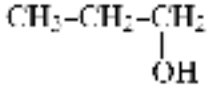
Hence,
Option (3) is correct : 1-Butene-3-yne

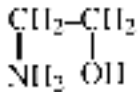
64) (1) 

(2) 

(3) 

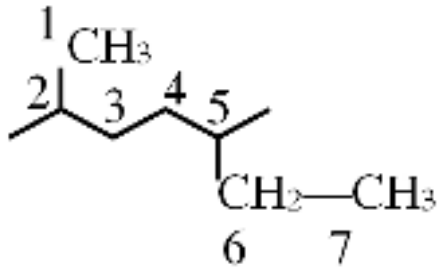
65)  ⇒ 1-methylcyclohexene





66)

Fact

67) 

68)

(A) is correct but (R) is not correct

69)

Explaining the Question :-
Evaluate the truth of an assertion about zero-order half-life and a reason about zero-order rate, and if the reason explains the assertion.
Concept:-
This question is based on Zero-order kinetics
Solution :-
Mathematical Analysis (in short):
Assertion: $t_{1/2} = \frac{[A]_0}{2k}$. Directly proportional to $[A]^0$. TRUE.
Reason: Rate = $k[A]^0 = k$. Independent of $[A]_0$. TRUE. Explanation: The reason explains the constant rate, not the direct proportionality of $t_{1/2}$ to $[A]_0$. The half-life dependence comes from the time it takes to consume half the initial amount at a constant rate.
Final Answer : (2)

70) **Question Explanation :** This is an assertion-reason type question testing understanding of pseudo first-order reactions.

Concept : Pseudo First-Order Reaction (acidic hydrolysis of ester)

Solution : Hydrolysis of ethyl acetate in acidic medium :
$CH_3COOC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$
The rate law is Rate = $k[CH_3COOC_2H_5][H_2O]$.
When water is in large excess, its concentration remains nearly constant, so the reaction appears to depend only on ester concentration — thus, pseudo first-order.
So, both assertion and reason are true, and reason correctly explains the assertion.

Final Answer : Option (4)

71)

ROD of $O_2 = \frac{8 \times 10^{-3}}{32} \text{mol L}^{-1}\text{S}^{-1}$
$-\frac{\Delta[O_2]}{\Delta t} = \frac{1}{2}\frac{-\Delta[SO_2]}{\Delta t}$

$\frac{8 \times 10^{-3}}{32} \times 2 =$ ROD of SO_2

ROD of $SO_2 = 5 \times 10^{-4} = 50 \times 10^{-5}$ mol L^{-1} S^{-1}

72) **Question Explanation:** Find the rate of appearance of B when the rate of disappearance of A is given for 2A → 3B + 4C.

Concept: This question is based on (relation between rate of disappearance and appearance using stoichiometry)

Solution:

Reaction: 2A → 3B + 4C

Rate relation: $-\frac{1}{2}\frac{d[A]}{dt} = \frac{1}{3}\frac{d[B]}{dt}$ Given: $-\frac{d[A]}{dt} = 4 \times 10^{-2} M/s \frac{1}{2} \times 4 \times 10^{-2} = \frac{1}{3}\frac{d[B]}{dt}$

$2 \times 10^{-2} = \frac{1}{3}\frac{d[B]}{dt} \frac{d[B]}{dt} =$ $3 \times 2 \times 10^{-2} = 6 \times 10^{-2}$ M/s **Final Answer :** Option (4)

73)

Question Explanation: Check order, rate law, units of k, and rate change.

Concept: Rate law: $r = k[A]^m[B]^n$ and units of k depend on overall order.

Solution:

Compare I → II: [A] doubles → rate doubles ⇒ order in A = 1

Compare II → III: [B] doubles → rate doubles ⇒ order in B = 1

⇒ Rate law: **r = k[A][B]** (b correct)

Now units of k:
Rate unit = mol L^{-1} s^{-1}
[A][B] unit = (mol L^{-1})^2

k unit = (mol L^{-1} s^{-1})/(mol^2 L^{-2}) = **L mol^{-1} s^{-1}**

Numerical value: k = 1, but unit ≠ s^{-1}
⇒ (a) incorrect

Doubling both concentrations:
Rate ∝ (2×2) = 4 times (c correct)

Final Answer: Option (3)

74)

Complex Reaction

75)

Theory

76) $t_{1/2} = \frac{A_0}{2K}; T_{100\%} = \frac{A_0}{K}$

77) **Explain Question:**
To find [B] at 100 min.

Concept:
Zero order Kinetics.

Solution:
Reaction : A —→ 2B
t = 0 1 -
t = 100 min 1–x 2x
$[A]_t = [A]_0 - kt \Rightarrow 1 - x = 1 - 10^{-3} \times 100$
$x = 0.1 \text{ mol } L^{-1}$

After 100 minutes ⇒ concentration of B = 2x = 0.2 mol L^{-1}

Final Answer: Option (2)

78)

Theory

79) $t = \frac{2.303}{K} \log \frac{A_0}{A_t}$

80)

1.6 M — 0.8 M — 0.4 M
Concentration after 2nd half time = 0.4 M
$r = K[A] = 2.31 \times 10^{-3} \times 0.4 = 9.2 \times 10^{-4} \text{ M } S^{-1}$

81) $K = \frac{1}{t} \ln \frac{100}{100-x}$

$K = \frac{1}{100} \ln 4$ .....(1)

$K = \frac{1}{t} \ln \frac{100}{10}$ .....(2)

From eq. (1) and (2)

$\frac{1}{100} \ln 4 = \frac{1}{t} \ln 10$

$t = \frac{100}{2 \times 0.3}$

82) **Explaining :** -
Concept : This question is based on
Solution : -

	A(g) →	B(g) +	C(g) +	D(g)
Initial	P_i	0	0	0
At time t	P_i–P	P	P	P

Total pressure at time t

$P_t = P_i - P + P + P + P \Rightarrow P_i + 2P$

$P_t = P_i + 2P$

$\frac{P_t - P_i}{2} = P$

1st order reaction

$$K = \frac{2.303}{t}\log\left(\frac{P_i}{P_i - P}\right)$$

$$K = \frac{2.303}{t}\log\left(\frac{P_i}{P_i - \frac{P_t - P_i}{2}}\right)$$

$$K = \frac{2.303}{t}\log\left(\frac{2P_i}{2P_i - P_t + P_i}\right)$$

$$K = \frac{2.303}{t}\log\left(\frac{2P_i}{3P_i - P_t}\right)$$

Final Answer : (3)

83) 10 minutes

84) $\frac{t_{1/8}}{t_{1/10}} = \frac{\frac{2.303}{K}\log\frac{a}{a/8}}{\frac{2.303}{K}\log\frac{a}{a/10}} = 0.9 \left[t = \frac{2.303}{K}\log\frac{A_0}{A_t}\right]$

85) **Explain Question:**

We are given a half-life expression: $t_{1/2} = \frac{\left(2-\sqrt{2}\right)}{K} \cdot C_0^{1/2}$ and asked to determine the order of reaction.

Concept: Half-life of nth order reaction

Solution:

For a reaction of order n, half-life is generally:

$t_{1/2} \propto \frac{1}{K} \cdot C_0^{1-n}$

Compare this with the given from:

$t_{1/2} \propto C_0^{1/2} \Rightarrow 1 - n = \frac{1}{2} \Rightarrow n = 0.5$

• The exponent of C_0 in half-life expression gives 1 – n = 0.5

Final Answer: Option (4)

86)

Explain Question:

We need to find order of reaction if half life and concentration of two samples of a substance are given.

Concept:

Relation between Half life and order of reaction.

Solution:

$t_{1/2} \propto \frac{1}{(a_0)^{n-1}}$ where n = order

$$\frac{0.1}{0.8} = \left(\frac{50}{400}\right)^{n-1}$$

$(1/8)^1 = (1/8)^{n-1}$

n – 1 = 1

$\boxed{n = 2}$

Final answer: option (2)

87) **Question Explanation:**

This question checks understanding of activation energy, molecular energy distribution, and the Arrhenius factor.

Concept:
This question is based on collision theory and Maxwell–Boltzmann energy distribution.

Solution:
(A) True — activation energy is the energy needed to form the activated complex.
(B) Area under the curve for the plot of fraction of molecules versus kinetic energy of gas molecules remains the same with temperature.

(C) In the Arrhenius equation the factor $e^{-Ea/RT}$ corresponds to fraction of molecules that have kinetic energy greater than activation energy.

(D) Activation energy of reaction does not depends upon temperature.

So B,C,D are incorrect.

Final Answer: Option 1

88)

Explain Question: We must match each each statement in Column I with the correct concept in Column II related to chemical kinetics.
Concept: Arrhenius Equation
Solution: Catalyst → lowers E_a
Molecularity → cannot be fractional or zero
First-order $t_1/_2$ → constant, independent of concentration
$e^{-E_a/RT}$ → fraction of molecules having $E \geqslant E_a$
Energetically favourable but slow → improper orientation
Area under MB curve → total probability = 1
A → R (catalyst lowers activation energy)

B →P (molecularity cannot be zero or fractional)
C → S (second half-life = first half-life for first-order reactions
D → U (fraction of molecules with E ⩾ threshold energy)
E →Q (slow due to improper orientation)
F → T (total probability = 1)
So the matching order: A → R, B → P,C → S,D → U,E →Q,F →T
This corresponds to option 1.
Final answer: option (1)

89)

$E_{af} - E_{ab} = \Delta H$(with sign)

90) E_a of 1st step must be higher than 2nd step and XY has lower energy than X + Y.

BIOLOGY

91)

NCERT Pg No # 60

92)

The correct sequence for the male genital tract is:

4. Tubuli recti → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Urethra

A. **Tubuli recti:** These are the straight tubules that connect the seminiferous tubules (where sperm are produced) to the rete testis.
B. **Rete testis:** A network of tubules that carries sperm from the seminiferous tubules to the efferent ducts.
C. **Vasa efferentia:** These are the efferent ducts that carry sperm from the rete testis to the epididymis.
D. **Epididymis:** A coiled tube where sperm mature and are stored.
E. **Vas deferens:** A long, muscular tube that carries sperm from the epididymis to the ejaculatory ducts.
F. **Urethra:** The final passageway through which sperm and urine are expelled from the body.

93) NCERT Pg. # 44

94) NCERT Pg. # 60

95)

NCERT Pg. 54

96) NCERT Pg. # 48(E), 53(H)

97)

NCERT Pg#33

98) NCERT Pg#38

99) NCERT Pg#30

100) NCERT Pg#29

101) NCERT Pg#28

102) NCERT Pg#33

103)

Module 7

104) NCERT Pg#36

105) NCERT Pg#38

106) NCERT Pg#38

107) NCERT Pg#37

108) Module 7

109) NCERT XII Pg. # 46

110)

NCERT-XII, Pg. 54

111) NCERT XII Page No. 31, 34

112) NCERT Page No. 38

113) NCERT Page No. 37

114)

NCERT-XII, Pg. # 35

115) **Explanation :**

A. **Fertilization (A)** occurs in the **fallopian tube (iii).**
B. **Implantation (B)** happens in the **uterus (i).**
C. **Sperm maturation (C)** takes place in the **epididymis (iv).**
D. **Oogenesis (D)** occurs in the **ovary (ii).**

The correct answer is option 4

116)

Option 3: C is the correct answer.

Explanation:

A. Structure C in the image represents a Graafian follicle.

B. Within the Graafian follicle, the developing egg cell (oocyte) is surrounded by a layer of cells called the cumulus oophorus.

C. The cumulus oophorus plays a crucial role in oocyte maturation and ovulation

117)

Epididymis can temporarily stores the sperm for as long as one month and here the functional maturation of sperms takes place.

118) NCERT Pg.# 250 (E), 273 (H)

119)

Option 4 is the mismatched pair.

A. The tail of the sperm contains a **flagellum**, which is responsible for its movement.
B. The energy for this movement is provided by **numerous mitochondria** located in the **midpiece** of the sperm, not the tail itself.

120) NCERT XIIth Pg.#48

121) NCERT-XII, Pg. # 36

122)

NCERT XII Pg. # 52 Fig. 3.11

123) NCERT Pg. # 34

124) NCERT Pg. # 33,34

125) NCERT-XII, Pg. # 27, Para-2.1

126) NCERT, Pg. # 27, 41, 46

127) NCERT, Pg. # 37

128) NCERT, Pg. # 35

129) Module No. # 8, Pg. No. # 23

130) Module No. # 8, Pg. No. # 43

131) NCERT Pg. # 28-38

132) NCERT, Pg. # 32

133) NCERT, Pg. # 36

134) NCERT, Pg. # 45,46

135)

NCERT-XII, Pg. # 26

136)

NCERT XI Pg. # 20

137) **Question Explanation:** Identifying the primary biological process associated with the specialized cell structure known as the heterocyst in Cyanobacteria.
Concept: Function of Heterocysts in Cyanobacteria
Solution: Heterocysts are specialized, thick-walled cells found in filamentous Cyanobacteria (Blue-Green Algae) like *Nostoc* and *Anabaena*. Their primary biological function is:

A. **N2 fixation:** They contain the oxygen-sensitive enzyme **nitrogenase**, which converts atmospheric nitrogen gas (N_2) into ammonia (NH_3). The heterocyst maintains the necessary **anaerobic environment** by lacking Photosystem II, which normally produces oxygen.

While the surrounding vegetative cells perform photosynthesis, the heterocyst itself is responsible for nitrogen fixation. They are not involved in sexual reproduction.
Final Answer: N_2 fixation

138)

Explain Question: Question asks which saprophytic protists form an aggregation called plasmodium.

Concept: This question is based on Slime moulds

Solution: Slime moulds are saprophytic protists; under favorable conditions they aggregate to form a multinucleate mass called plasmodium.
Final answer: 3

139)

Explain Question: Question asks the cause of red tide.

Concept: This question is based on Dinoflagellates

Solution: Red tide is due to rapid multiplication of dinoflagellates (Gonyaulax)
Final answer: 4

140)

Explain Question: Question asks how many of the listed groups are photosynthetic.

Concept: This question is based on Protista

Solution: 3 groups are photosynthetic (Dinoflagellates, Diatoms, Euglenoids).
Final answer: 2

141) **Question Explanation**: Reason for dinoflagellate nickname.
Concept: Dinoflagellate Cell Wall.
Solution: Dinoflagellates are called "**armoured algae**" because their outer surface is covered by rigid, interlocked plates made of **cellulose**. This structure forms a protective layer, giving them an armored appearance.
Final Answer: Their cell wall is made up of plates

142) **Explaining the Question :-** Question asks about chemoautotrophic bacteria in nitrogen metabolism.
Concept:- This question is based on Nutrition in bacteria
Solution :- Nitrifying bacteria are chemoautotrophs
Nitrosomonas oxidizes $NH_4^+ \rightarrow NO_2^-$
Nitrobacter oxidized $NO_2^- \rightarrow NO_3^-$

Mycobacterium (pathogen) Rhodospirillum (photosynthetic) clostridium (anaerobic N_2 fixer) are not chemoautotrophic nitrifiers.
Final Answer :- 2

143) **Explaining the Question :-** Question asks which statements are correct/incorrect.

Concept:- This question is based on structure of bacterial cell
Solution :- (A) Extra-genomic DNA (plasmid) provides antibiotic resistance. -True
(B) Mesosomes help in respiration & DNA replication -True
(C) pili and fimbriae are for attachment & conjugation, not motility - False

(D) Glycocalyx + cell wall + cell membrane = cell envelope - True
Final Answer :- 1

144)

NCERT XI(E)/(H) Pg # 19 (2.1.2)

145)

Explaining the Question :- Question asks which statements are correct.

Concept:- This question is based on Monera

Solution :- (A) Methanogens (archaebacteria) produce methane in marshy places.-True
(B) Nostoc (cyanobacterium) fixes atmospheric N_2 in heterocysts.-True
(C) Chemosynthetic bacteria oxidize inorganic substances (NH_3, H_3S, Fe^{2+}), not form cellulose.-False
(D) Mycoplasma is wall-less, anaerobic or facultatively anaerobic, can survive without oxygen-True
Final Answer :- 4

146)

Explain Question : The question asks to identify the organisms shown (A and B) and choose correct statements about them
Concept : This question is based on PROTISTA AND FUNGI
Explanation : A = Euglena (unicellular protist, both autotrophic & heterotrophic, motile)
B = Agaricus (mushroom, eukaryotic, heterotrophic, non-motile)
Both are eukaryotes, but only Euglena is motile.
Final answer :3

147)

Explain Question : The question is asking us to identify the figure and the labelled parts (a, b, c).

Concept : This question is based on Reproduction in bacteria.

Solution :

The figure shows binary fission in bacteria.

In bacteria:
a = **DNA** (nucleoid region).
b = **Cell membrane** (plasma membrane).
c = **Cell wall** (outer layer).

Final answer : option (2)

148)

Question Explanation: Question asks about identifying the labelled parts A and B in the diagram of filamentous blue-green alga Nostoc.

Concept: Cyanobacteria

Solution: A = Heterocyst

B = Mucilaginous sheath

Final answer: 1

149)

Question Explanation : The question asks about identification of labelled A and B in given digram and select correct answer from given options.

Concept: Morphology of Bacteria

Solution: A is rod shaped bacteria so it will be bacillus and B is endospore

Final answer: 2 A-Bacilli, B-Spore

150) **Explain Question:** Question asks the causal organism for mushroom, bread mould, ringworm, and white rust.
Concept: This question is based on Fungi general characters
Solution: Common mushroom → Agaricus (Fungus)
Bread mould → Rhizopus (Fungus)
Ringworm → Fungus (dermatophyte)
White spots on mustard (Albugo) → Fungus (Phycomycetes)
All are caused by fungi.
Final answer: 3

151) **Explain Question:** Question asks to identify the correct match from the three columns.
Concept: This question is based on Fungal classes
Solution: Albugo → Phycomycetes, obligate parasite → (1-c-i).
Agaricus → Basidiomycetes, edible mushroom → (2-d-iv).
Aspergillus → Ascomycetes, lab weed → (3-a-iii).

Alternaria → Deuteromycetes, decomposer of litter → (4-b-ii).
Final answer: 4

152)

Explain Question: Question asks to choose the incorrect match.

Concept: This question is based on Fungal classes

Solution: Incorrect match = Agaricus – Ascomycetes. Bcz it belong from basidiomycetes

Final answer: 2

153)

Explain Question: Question asks to find the incorrect statement.

Concept: This question is based on Protista

Solution: (1) Chrysophytes (diatoms, desmids) occur in fresh & marine water.-T

(2) Dinoflagellates are mostly marine, not freshwater.-False

(3) Most euglenoids are freshwater.-T

(4) Slime moulds are saprophytic protists.-T

Final answer: 2

154)

Explain Question: Question asks which statements are true for archaebacteria among the given options.

Concept: Archaebacteria

Solution: Archaebacteria include —
• Methanogens (produce methane) , • Extreme halophiles (live in high salt)
• Thermoacidophiles (live in hot, acidic environments) , .Their cell wall lacks peptidoglycan.

Final answer: (4)

155)

NCERT-XI, Pg. # 128, 129

156) NCERT (XI) Pg. # 19

157) **Explain Question:** Question asks in which organisms heterocysts are found.

Concept: Cyanobacteria
Solution: Heterocysts occur in Anabaena and Nostoc.
Final answer: 1

158)

Explain Question : The question asks to identify the specific type of microorganisms responsible for producing methane gas in the digestive system (gut) of ruminant animals.

Concept : This question is based on Archaebacteria.

Solution : "Methanogens are a group of archaebacteria that produce methane in the guts of ruminants."

Final Answer : option (4)

159)

NCERT XI Pg. # 21,32

160)

Explain Question:
The question asks to correctly match the kingdoms (Monera, Protista, Plantae, Fungi) with their representative organisms.

Concept:
Kingdoms (Monera, Protista, Plantae, Fungi)

Solution:
(A) Monera → (II) Bacillus, Nostoc
(B) Protista → (III) Euglena, Trypanosoma
(C) Plantae → (I) Spirogyra, Solanum
(D) Fungi → (IV) Mucor, Rhizopus

Final answer: Option (2)

161)

NCERT XI Pg. # 20

162)

NCERT XI Pg. # 20

163)

NCERT Pg. # 24, [2.3.4]

164)

Explain Question: Question asks about evaluating the truth of the assertion and reason regarding why viruses and viroids are not included in Whittaker's five-kingdom classification.

Concept: Whittaker's five-kingdom classification

Solution: Assertion: Viruses and viroids are not mentioned → True
Reason: They are not truly living → True
The reason correctly explains the assertion.

Final answer: (1)

165)

NCERT Pg # 23

166)

Question Explanation: Carl Woese reorganized life into three domains by naturally clustering kingdom based on rRNA differences.

Concept: Carl Woese Classification

Solution: Woese's domains:- Two domains for prokaryotes (Bacteria, Archaea) One domain for eukaryotes (Eukarya)

Final Answer: option (2)

167) **Question Explanation**: Identifying correct facts about Deuteromycetes.
Concept: Characteristics of Imperfect Fungi.
Solution: (a) **True**: Many members are vital **decomposers** of litter, aiding mineral cycling. (b) **True**: They are commonly known as **Imperfect Fungi** because their sexual stage is unknown or absent. (c) **False**: While some are parasites, **most** are **decomposers** (saprophytes), not parasites. (d) **False**: Their defining feature is that the **sexual phase is not known**.Therefore, statements (a) and (b) are true.
Final Answer: a & b

168)

Question Explanation :
Question is asking about correct statement for 5 kingdom system.

Solution :

A. (A) Correct → Monera, Plantae, and Protista contain autotrophic organisms.
B. (B) Incorrect → Three kingdoms (Plantae, Animalia, and Fungi) are multicellular, not just two.
C. (C) Correct → Four kingdoms (Protista, Fungi, Plantae, Animalia) have a nuclear membrane, while Monera lacks it.
D. (D) Incorrect → Only Monera(some bacteria) contain chemosynthetic autotrophs, not all

kingdoms.

Final Answer :
Option (3)

169)

A. Statement B:
Ascospores are typically 8 in number (sometimes 4) and are formed inside a sac-like structure called an ascus (endogenously formed). Thus, this statement is correct.
B. Statement D:
Nostoc and Anabaena are indeed nitrogen-fixing cyanobacteria (autotrophic bacteria) often found in the coralloid roots of cycads, forming a symbiotic relationship. Hence, this statement is correct.

The correct answer is:OPTION(4) B&D

170)

Explain Question: Question asks whether both statements about Mycoplasma are true or false.

Concept: Mycoplasma

Solution: I. Mycoplasma are the smallest living cells and can survive without oxygen (facultative anaerobes)-CORRECT
II. Many Mycoplasma species are pathogenic to animals and plants-CORRECT

Final answer: 4

171)

NCERT XI Pg no. 15

172)

Question Explanation: Analyzing statements about Archaebacteria characteristics.

Concept: Archaebacteria.

Solution:

Statement-I : Archaebacteria have complex cell wall and complex cell membrane. **Statement-II :** Archaebacteria can survive in extremely harsh habitats.

Both Statement I and Statement II are correct.

So, the correct answer is 4.

Final Answer: Option (4)

173)

Explain Question : Question is asking which group of organisms is described by the given statements about plasmodium, spore formation, and dispersal.

Concept : This question is based on slime moulds

Solution : The characteristics listed (plasmodium → fruiting bodies, resistant airborne spores) belong to slime moulds.

Final answer : option (2)

174)

NCERT XI Pg. # 16

175) **Explain Question:** Question asks about which statements are incorrect regarding bacteria.
Concept: This question is based on Bacteria
Solution: (A) Bacterial structure is simple but behaviour is complex → Correct.
(B) Bacteria show most extensive metabolic diversity → Correct.C) Vast majority are autotrophs and a few heterotrophs → Incorrect, actually most are heterotrophs, only some are autotrophs.
(D) Cell wall made up of lipoprotein → Incorrect, bacterial cell wall is made of peptidoglycan (murein), not lipoprotein.
Incorrect = C & D.
Final answer: 2

176) **Explain Question:** Question asks how many statements are incorrect.
Concept: This question is based on Bacteria and Protista
Solution: (A) True → In five kingdom classification, unicellular algae like Chlamydomonas & Chlorella are in Protista.
(B) False → Entamoeba coli is a protozoan, not a bacterium.
(C) False → Paramecium & Plasmodium (Protista) and Penicillium (Fungi).
(D) True → Purple sulphur bacteria use H_2S as hydrogen donor.
Final answer: 3

177)

NCERT XI Pg # 19, 20

178) *Nostoc, Azotobacter, Rhodospirillum, Beijernickia, Anabaena, Frankia*

179)

NCERT-XI Pg. # 15

180) **Explain Question:** Question asks about how many statements are true for fungi among the given six statements.

Concept: General characters of fungi and fungal classes

Solution: Check each statement:

(A) True → Fungi show huge diversity.
(B) True → Majority are saprophytes.
(C) False → Asexual reproduction does NOT involve three steps.
(D) False → Morels & truffles are edible delicacies but not mushroom
(E) False → Albugo infects mustard only, not wheat.
(F) True → Ascomycetes include yeast (unicellular) + molds (multicellular).

True statements = A, B, , F → Total = 3

Final answer: 3